Chapter 1
Variational Formulations and Finite Element Methods

Although we shall not define in this chapter mixed and hybrid (or other non-standard) finite element methods in a very precise way, we would like to situate them in a sufficiently clear setting. As we shall see, boundaries between different methods are sometimes rather fuzzy. This will not be a real drawback if we nevertheless know how to apply correctly the principles underlying their analysis.

After having briefly recalled some basic facts about classical methods, we shall present a few model problems. The study of these problems will be the kernel of this book. Thereafter, we rapidly recall basic principles of *duality theory* as this will be our starting point to introduce mixed methods. *Domain decomposition* methods (allied to duality) will lead us to hybrid methods. Then we shall briefly discuss modified variational formulations that can be used to obtain better stability properties for the discretised versions.

1.1 Classical Methods

We recall here, in a very simplified way, some facts about optimisation methods and the classical finite element method. Such an introduction cannot be complete and does not want to be. We refer the reader to [146] or [334], among others, where standard finite element methods are clearly exposed. We also refer to [167] where an exhaustive analysis of many of our model problems can be found.

Let us consider a very common situation where the solution of a physical problem minimises some functional (usually an "energy functional"), in a "well chosen" space of admissible functions V that we take for the moment as a Hilbert space:

$$\inf_{v \in V} J(v). \tag{1.1.1}$$

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If the functional $J(\cdot)$ is differentiable (cf. [184] for instance) the minimum (whenever it exists) will be characterised by *a variational equation*

$$\langle J'(u), v\rangle_{V'\times V} = 0, \quad \forall v \in V, \tag{1.1.2}$$

where $\langle\cdot,\cdot\rangle_{V'\times V}$ denotes duality between V and its topological dual V', the derivative $J'(u)$ at point u being considered as a linear form on V.

The classical Ritz's method to approximate the solution of (1.1.1) consists in choosing a finite dimensional subspace V_m of V, and then looking for $u_m \in V_m$ solution of the problem

$$\inf_{v_m\in V_m} J(v_m), \tag{1.1.3}$$

or, differentiating,

$$\langle J'(u_m), v_m\rangle_{V'\times V} = 0, \quad \forall v_m \in V_m. \tag{1.1.4}$$

Let us consider, to fix ideas, a quadratic functional

$$J(v) := \frac{1}{2}a(v,v) - L(v), \tag{1.1.5}$$

where $a(\cdot,\cdot)$ is a bilinear form on V, which we suppose continuous and symmetric, and $L(\cdot)$ a linear form on V. The variational equation (1.1.2) can then be written as

$$a(u,v) = L(v) \quad \forall v \in V, \tag{1.1.6}$$

while the discrete problem (1.1.4) becomes

$$a(u_m, v_m) = L(v_m), \quad \forall\, v_m \in V_m,\ \ u_m \in V_m. \tag{1.1.7}$$

If a basis $w_1, w_2, \ldots, w_m$ of V_m is chosen, and if we write

$$u_m = \sum_{i=1}^{m} \alpha_i w_i, \tag{1.1.8}$$

problem (1.1.7) is reduced to the solution of the linear system

$$\sum_{i=1}^{m} a_{ij}\alpha_i = b_j, \ \ 1 \le j \le m, \tag{1.1.9}$$

where we set

$$a_{ij} := a(w_i, w_j), \ \ b_j := L(w_j). \tag{1.1.10}$$

This formulation can be extended to the case where the bilinear form $a(\cdot,\cdot)$ *is not symmetric* and where problem (1.1.7) no longer corresponds to a minimisation

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problem. This is then usually called a Galerkin method. Let us recall that problems of type (1.1.7) will have a unique solution if, in particular, the bilinear form $a(\cdot,\cdot)$ is *coercive*, that is if there exists a positive real number α such that for all v in V

$$a(v,v) \geq \alpha \|v\|_V^2. \tag{1.1.11}$$

The above described methodology is very general and classical. We can consider the finite element method as a special case in the following sense.

The finite element method is a general technique to build finite dimensional subspaces of a Hilbert space V in order to apply the Ritz-Galerkin method to a variational problem.

This technique is based on a few simple ideas. The fundamental one is the *partition of the domain Ω* in which the problem is posed, into a set of "simple" sub-domains, called elements. These elements are usually triangles, quadrilaterals, tetrahedra, etc. A space *V of functions defined on Ω is then approximated* by "simple" functions, defined on each sub-domain with suitable matching conditions at interfaces. Simple functions are usually polynomials or functions obtained from polynomials by a change of variables.

This, of course, a very summarised way of defining finite elements and this is surely not the best way to understand it from the computational point of view. We shall come back to this in Chap. 2 with a much more workable approach.

The point that we want to emphasise here is the following. *A finite element method can only be considered in relation with a variational principle and a functional space. Changing the variational principle and the space in which it is posed leads to a different finite element approximation (even if the solution for the continuous problems can remain the same).*

In the remaining of this Chapter, we shall see how different variational formulations can be built for the same physical problem. Each of these formulations will lead to a new setting for finite element approximations. *The common point of the methods analysed in this book is that they are founded on a variational principle expressing an equilibrium* (saddle point) *condition* rather than on a minimisation principle. We shall now try to see, on some examples, how such equilibrium principles can be built.

1.2 Model Problems and Elementary Properties of Some Functional Spaces

The aim of this section is to introduce some notation and to present five model problems that will underlie almost all cases analysed in this book. They will be the Dirichlet problem for Laplace's equation, the linear elasticity problem, Stokes' problem, a fourth-order problem modelling the deflection of a thin clamped plate, and, finally, the time-harmonic Maxwell system. These problems are closely interrelated and methods to analyse them will also be.

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We shall present, in this section, the most classical variational formulation of these problems. The following sections will lead us to less standard forms.

We shall assume, in our exposition, that the problems are posed in a domain Ω of $\mathbb{R}^n$, with a sufficiently smooth boundary $\partial\Omega = \Gamma$ (for instance a Lipschitz continuous boundary). In practice $n = 2$ or 3 and we shall present most of our examples in a two-dimensional setting for the sake of simplicity.

In the problems considered here, working in $\mathbb{R}^2$ rather than in $\mathbb{R}^3$ is not really restrictive and extensions are generally straightforward. (This is however not always the case for numerical methods.) Let us first recall some definitions. We shall constantly use Sobolev spaces [3, 281, 309]. They are based on

$$L^2(\Omega) := \left\{ v \Big| \int_\Omega |v|^2 \, dx = \|v\|^2_{L^2(\Omega)} < +\infty \right\}, \tag{1.2.1}$$

the space of square integrable functions on Ω. To be precise, instead of *functions* we should actually say *classes of measurable functions*, meaning that a class is made of functions that differ from each other only on a subset of Ω of zero Lebesgue measure. Having said this once, we shall keep calling them simply *functions*. We then define in general, for m integer ≥ 0,

$$H^m(\Omega) := \left\{ v \Big| \ D^\alpha v \in L^2(\Omega), \quad \forall |\alpha| \leq m \right\}, \tag{1.2.2}$$

where

$$D^\alpha v := \frac{\partial^{|\alpha|} v}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}, \qquad |\alpha| = \alpha_1 + \cdots + \alpha_n,$$

these derivatives being taken *in the sense of distributions*. On this space, we shall use the semi-norms

$$|v|^2_{k,\Omega} := \sum_{|\alpha|=k} |D^\alpha v|^2_{L^2(\Omega)}, \qquad k = 0, 1, \ldots, m, \tag{1.2.3}$$

and the norm

$$\|v\|^2_{m,\Omega} := \sum_{k \leq m} |v|^2_{k,\Omega}. \tag{1.2.4}$$

Remark 1.2.1. The norm in (1.2.4) is definitely *the weirdest* aspect of the whole theory of Sobolev spaces. Indeed, one should take a *typical length* ℓ of the problem (as for instance the diameter of Ω) and use, instead of (1.2.4):

$$\|v\|^2_{m,\Omega} := \sum_{k \leq m} \ell^{2k} |v|^2_{k,\Omega}, \tag{1.2.5}$$

avoiding, in this way, to sum objects with different physical dimensions. The expression (1.2.4), which is by far *the most widely used in all the international literature* assumes implicitly that the problem has been adimensionalised, something that one

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is not always willing to do. Needless to say, (1.2.4) has a lot of advantages, and we are often going to use it. Nevertheless, we felt compelled to give at least a minor warning to our readers. □

The space $L^2(\Omega)$ is then $H^0(\Omega)$ and we shall usually write $\|v\|_{0,\Omega}$, to denote its norm $\|v\|_{L^2(\Omega)}$. Let us denote as usual by $\mathfrak{D}(\Omega)$ the space of infinitely differentiable functions having a compact support in Ω. We denote by $H_0^m(\Omega)$ the completion of $\mathfrak{D}(\Omega)$ for the topology defined by the norm (1.2.4). If the boundary is smooth enough (e.g. Lipschitz continuous boundary) this simple definition will coincide, without troublesome pathologies, with the more common

$$H_0^m(\Omega) := \left\{v|\ v \in H^m(\Omega) \text{ s. t. } v = \frac{\partial v}{\partial n} = \cdots = \frac{\partial^{m-1} v}{\partial n^{m-1}} = 0 \text{ on } \Gamma\right\}, \quad (1.2.6)$$

where $\underline{n}$ is the normal direction to $\Gamma = \partial\Omega$. The drawback in the definition (1.2.6) is however the difficulty in giving sense to the value of v (and, if $m > 1$, to its derivatives) on the boundary of Ω. We shall shortly give a hint on how this could be made precise for the most common cases $m = 1$ and $m = 2$.

Indeed, among the spaces introduced so far, the most commonly used, apart from $L^2(\Omega)$, will be $H^1(\Omega)$, $H_0^1(\Omega)$, $H^2(\Omega)$ and $H_0^2(\Omega)$.

If the boundary $\partial\Omega$ is sufficiently smooth (and, again, Lipschitz continuity will be enough), one can show that there exists a linear and continuous operator $\gamma_0 : H^1(\Omega) \to L^2(\Gamma)$ such that $\gamma_0 v$ *coincides with the restriction of* v *to* Γ *whenever* v *is smooth* (say, to fix the ideas, for every $v \in C^1(\bar{\Omega})$). It seems then natural to call $\gamma_0 v$ "the trace of v on Γ" and denote it by $v|_\Gamma$ even if v is a general function in $H^1(\Omega)$ that might not be in $C^1(\bar{\Omega})$.

A deeper analysis shows that by taking all the traces of all the functions of $H^1(\Omega)$, one does not obtain the whole space $L^2(\Gamma)$ but only a subspace of it. Further investigations show that such a subspace contains $H^1(\Gamma)$ as a proper subset. Hence we have,

$$H^1(\Gamma) \subset \gamma_0(H^1(\Omega)) \subset L^2(\Gamma) \equiv H^0(\Gamma), \quad (1.2.7)$$

where every inclusion is strict. It is finally recognised that the space $\gamma_0(H^1(\Omega))$ belongs to a family of spaces $H^s(\Gamma)$ (defined for all $s \in \mathbb{R}$, that we are not going to detail here) and corresponds exactly to the value $s = \frac{1}{2}$. Hence we have

$$H^{\frac{1}{2}}(\Gamma) := \gamma_0(H^1(\Omega)), \quad (1.2.8)$$

with

$$\|g\|_{H^{\frac{1}{2}}(\Gamma)} := \inf_{\substack{v \in H^1(\Omega) \\ \gamma_0 v = g}} \|v\|_{H^1(\Omega)}. \quad (1.2.9)$$

In a similar way we could see that the traces of functions in $H^2(\Omega)$ belong to a space $H^s(\Gamma)$ for $s = \frac{3}{2}$. We may therefore set

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$$H^{\frac{3}{2}}(\Gamma) := \gamma_0(H^2(\Omega)), \tag{1.2.10}$$

$$\|g\|_{H^{\frac{3}{2}}(\Gamma)} := \inf_{\substack{v\in H^2(\Omega)\\ \gamma_0 v = g}} \|v\|_{H^2(\Omega)}. \tag{1.2.11}$$

This can be generalised to the traces of higher order derivatives. For instance, if the boundary Γ is Lipschitz continuous, one can define a linear continuous operator $\gamma_1 : H^2(\Omega) \to L^2(\Omega)$ such that $\gamma_1 v$ coincides with the trace of the normal derivative of v whenever v is smooth (say, $v \in C^2(\bar{\Omega})$). Proceeding as before we could then define, for $v \in H^2(\Omega)$, the *trace of the normal derivative*:

$$\frac{\partial v}{\partial n}\Big|_\Gamma \equiv \gamma_1 v.$$

If the boundary Γ is smooth enough (but, here, Lipschitz continuity will *not* be enough: one needs at least C^1) one gets

$$\gamma_1(H^2(\Omega)) = H^{\frac{1}{2}}(\Gamma) \qquad \text{(for } \partial\Omega \text{ smoother: not for a polygon)}.$$

Note however that for less regular domains this will no longer be true: for instance, if Ω is a polygon we have that $\gamma_1 v$ belongs to a more complicated space that, roughly speaking, is made of functions whose restriction to each edge e belongs to $H^{\frac{1}{2}}(e)$. We shall not discuss in a more precise way trace theorems on Sobolev spaces of fractional order. (The reader may refer to the authors quoted above.) Intuitively, Sobolev spaces of fractional order can be considered as having regularity properties that are intermediate between the properties of the neighbouring integer order spaces and they can indeed be defined as *interpolation spaces*. Taking this as granted, we then have

$$H_0^1(\Omega) := \left\{v|\ v \in H^1(\Omega),\ \ v\big|_\Gamma := 0\right\}, \tag{1.2.12}$$

$$H_0^2(\Omega) = \left\{ v|\ v \in H^2(\Omega),\ v\big|_\Gamma = 0,\ \frac{\partial v}{\partial n}\Big|_\Gamma = 0 \right\}. \tag{1.2.13}$$

For $v \in H_0^1(\Omega)$, we have the *Poincaré inequality*

$$\|v\|_{0,\Omega} \le C(\Omega)|v|_{1,\Omega}, \tag{1.2.14}$$

and the semi-norm $|\cdot|_{1,\Omega}$ is therefore a norm on $H_0^1(\Omega)$, equivalent to $\|\cdot\|_{1,\Omega}$. We shall also need to consider functions that vanish on a part of the boundary; suppose that $\Gamma = D \cup N$ is a "reasonable" partition of Γ into disjoint parts; then we can define

$$H_{0,D}^1(\Omega) := \{v|\ v \in H^1(\Omega),\ \ v\big|_D = 0\}, \tag{1.2.15}$$

and one has $H_0^1(\Omega) \subset H_{0,D}^1(\Omega) \subset H^1(\Omega)$.

When considering vector-valued functions, additional spaces will be useful. In particular we shall use

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$$H(\mathrm{div};\Omega) := \{\underline{v} \in L^2(\Omega)^n \,|\, \mathrm{div}\,\underline{v} \in L^2(\Omega)\}, \tag{1.2.16}$$

$$H(\underline{\mathrm{curl}};\Omega) := \{\underline{v} \in L^2(\Omega)^n \,|\, \underline{\mathrm{curl}}\,\underline{v} \in L^2(\Omega)^d\}, \tag{1.2.17}$$

where in (1.2.17) we have $d = 1$ when $n = 2$ and $d = 3$ when $n = 3$. The divergence and curl operators are defined as usual; particular care has to be taken for the definition of the two-dimensional curl operator which in this case is understood as $\underline{\mathrm{curl}}\,\underline{v} = \mathrm{div}(\underline{v}^\perp)$, where $\underline{v}^\perp$ is the rotation of $\underline{v}$ by an angle of $\pi/2$. It can be shown that functions in $H(\mathrm{div};\Omega)$ (resp. $H(\underline{\mathrm{curl}};\Omega)$) admit traces of the normal (resp. tangential) component on Γ; namely, there exists a linear and continuous operator $\gamma_n : H(\mathrm{div};\Omega) \to H^{-1/2}(\Gamma)$ such that $\gamma_n \underline{v} =$ trace of $\underline{v} \cdot \underline{n}$ for every smooth $\underline{v}$, where $\underline{n}$ is the outward normal unit vector, and a similar property holds for the traces of the tangential components of vectors in $H(\underline{\mathrm{curl}};\Omega)$. More details on $H(\mathrm{div};\Omega)$ and $H(\underline{\mathrm{curl}};\Omega)$ will be given in Chap. 2. Spaces including homogeneous boundary conditions are denoted as follows

$$H_0(\mathrm{div};\Omega) := \{\underline{v} \mid \underline{v} \in H(\mathrm{div};\Omega),\ \underline{v} \cdot \underline{n} = 0 \text{ on } \Gamma\}, \tag{1.2.18}$$

$$H_0(\underline{\mathrm{curl}};\Omega) := \{\underline{v} \mid \underline{v} \in H(\underline{\mathrm{curl}};\Omega),\ \underline{v} \times \underline{n} = 0 \text{ on } \Gamma\}. \tag{1.2.19}$$

We shall come back in Chap. 2 to the properties of these spaces; the above definitions are sufficient to allow us to present some examples.

Example 1.2.1 (Boundary value problems for the Laplace equation). This is a very classical case that in fact led to the definition of Sobolev spaces. For f given in $L^2(\Omega)$, let us consider the following minimisation problem on $H_0^1(\Omega)$:

$$\inf_{q \in H_0^1(\Omega)} \Big(\frac{1}{2} \int_\Omega |\,\underline{\mathrm{grad}}\, q|^2 \, dx - \int_\Omega f q \, dx \Big), \tag{1.2.20}$$

where $|\,\underline{\mathrm{grad}}\, q|^2 = \big|\frac{\partial q}{\partial x_1}\big|^2 + \big|\frac{\partial q}{\partial x_2}\big|^2 = \underline{\mathrm{grad}}\, q \cdot \underline{\mathrm{grad}}\, q$. One shows easily (cf. [141, 281, 309] for instance) that this problem has a unique solution p, characterised by: $p \in H_0^1(\Omega)$ and

$$\int_\Omega \underline{\mathrm{grad}}\, p \cdot \underline{\mathrm{grad}}\, q \, dx = \int_\Omega f q \, dx, \quad \forall q \in H_0^1(\Omega). \tag{1.2.21}$$

This is of the form (1.1.6), by setting

$$a(p,q) = \int_\Omega \underline{\mathrm{grad}}\, p \cdot \underline{\mathrm{grad}}\, q \, dx$$

This solution p satisfies, in the sense of distributions,

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$$\begin{cases} -\Delta p = f \text{ in } \Omega, \\ p\big|_{\Gamma} = 0, \end{cases} \tag{1.2.22}$$

which is a standard Dirichlet problem for the Laplace operator Δ, commonly called *Poisson problem*. If $H_0^1(\Omega)$ were replaced by $H_{0,D}^1(\Omega)$ one would get instead of (1.2.22) a mixed type problem

$$\begin{cases} -\Delta p = f \text{ in } \Omega, \\ p = 0 \text{ on } \Gamma_D, \\ \dfrac{\partial p}{\partial n} = 0 \text{ on } \Gamma_N. \end{cases} \tag{1.2.23}$$

We thus have Dirichlet boundary conditions on Γ_D and Neumann conditions on Γ_N. In particular for $\Gamma_N = \Gamma$, we get a Neumann problem. It must be noted that minimising (1.2.20) on $H^1(\Omega)$ instead of $H_0^1(\Omega)$ will define p up to an additive constant and requires the compatibility condition

$$\int_{\Omega} f \, dx = 0,$$

which can be seen to be necessary from (1.2.21) by taking $q \equiv 1$ in Ω.

If we denote by $H^{-\frac{1}{2}}(\Gamma)$ the dual space of $H^{\frac{1}{2}}(\Gamma)$, and we take $g \in H^{-\frac{1}{2}}(\Gamma)$, we can consider the functional

$$\frac{1}{2}\int_{\Omega} |\operatorname{\underline{grad}} q|^2 \, dx - \int_{\Omega} fq \, dx + \langle g, q \rangle, \tag{1.2.24}$$

where the bracket $\langle \cdot, \cdot \rangle$ denotes the duality between $H^{-\frac{1}{2}}(\Gamma)$ and $H^{\frac{1}{2}}(\Gamma)$. We shall sometimes write *formally* $\int_{\Gamma} gq \, ds$ instead of $\langle g, v \rangle$. Minimising (1.2.24) on $H_{0,D}^1(\Omega)$ leads to the problem

$$\begin{cases} -\Delta p = f \text{ in } \Omega, \\ p = 0 \text{ on } \Gamma_D, \\ \dfrac{\partial p}{\partial n} = g \text{ on } \Gamma_N. \end{cases} \tag{1.2.25}$$

When $\Gamma_D = \emptyset$ the solution is defined up to an additive constant and we must choose f and g such that

$$\int_{\Omega} f \, dx - \int_{\Gamma} g \, ds = 0.$$

These problems are among the most classical of mathematical physics and we do not have to emphasise their importance. In the following chapters we shall need to use *regularity results* for the problems introduced above. We have supposed up to now $f \in L^2(\Omega)$. For the Poisson problem (1.2.22) we could have assumed f to

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belong to a weaker space, namely $f \in H^{-1}(\Omega) = (H_0^1(\Omega))'$, and nevertheless obtained $p \in H_0^1(\Omega)$. On the other hand if f is taken in $L^2(\Omega)$ and Ω is convex one can prove [309] that $p \in H^2(\Omega)$ and that

$$\|p\|_{2,\Omega} \leq c\|f\|_{0,\Omega}, \tag{1.2.26}$$

where c is a constant depending only on Ω. Regularity results are essential to many approximations results and are fundamental to obtain error estimates. We refer the reader to [233] for the delicate questions of the regularity of the general problem (1.2.23) in a domain with corners. □

Example 1.2.2 (Linear elasticity). We want to determine the displacement $\underline{u} = \{u_1, u_2\}$ of an elastic material under the action of some external forces. We suppose the displacement to be small and the material to be isotropic and homogeneous [146, 295]. The domain Ω is the initial configuration of the body. To set our problem, we must introduce some notation from continuum mechanics. First we define the linearised strain tensor $\underline{\underline{\varepsilon}}(\underline{u})$ by

$$\underline{\underline{\varepsilon}}(\underline{u}) := \frac{1}{2}\Big(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\Big). \tag{1.2.27}$$

The trace, $\mathrm{tr}(\underline{\underline{\varepsilon}})$, of this tensor is nothing but the divergence of the displacement field

$$\mathrm{tr}(\underline{\underline{\varepsilon}})(\underline{v}) = \mathrm{div}\ \underline{v}. \tag{1.2.28}$$

We shall also use the *deviatoric* $\underline{\underline{\varepsilon}}^D$ of the tensor $\underline{\underline{\varepsilon}}$ that is

$$\underline{\underline{\varepsilon}}^D := \underline{\underline{\varepsilon}} - \frac{1}{n}\,\mathrm{tr}(\underline{\underline{\varepsilon}})\underline{\underline{\delta}}, \tag{1.2.29}$$

where $\underline{\underline{\delta}}$ is the standard Kronecker tensor and n is the space dimension. The deviatoric is evidently built to have $\mathrm{tr}(\underline{\underline{\varepsilon}}^D) = 0$. Let then Γ_0 be a part of Γ on which we assume $\underline{u} = 0$. We also assume the existence in Ω of a distributed force $\underline{f}$ (e.g. gravity) and on Γ_1 of a traction $\underline{g}$ that is decomposed into a normal part g_n and a tangential part g_t. We denote by $\underline{n}$ and $\underline{t}$ the normal and tangential unit vectors to Γ. Let us denote

$$|\underline{\underline{\varepsilon}}|^2 = \underline{\underline{\varepsilon}} : \underline{\underline{\varepsilon}} := \sum_{i,j} \varepsilon_{ij}^2 \tag{1.2.30}$$

and let us consider in $V := (H^1_{0,\Gamma_0}(\Omega))^2$ the minimisation problem

$$\begin{aligned}\inf_{\underline{v}\in V}\Big\{ \int_\Omega \frac{1}{2}\big(\lambda|\ \mathrm{div}\ \underline{v}|^2 + 2\mu|\underline{\underline{\varepsilon}}(v)|^2\big)\,dx - \int_\Omega \underline{f}\cdot\underline{v}\,dx \\ - \int_{\Gamma_1} g_n \underline{v}\cdot\underline{n}\,ds - \int_{\Gamma_1} g_t\,\underline{v}\cdot\underline{t}\,ds\Big\}.\end{aligned} \tag{1.2.31}$$

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Constants λ and μ, the Lamé coefficients, depend on the physical properties of the material considered. The solution $\underline{u}$ of this problem is then characterised by

$$\begin{aligned} 2\mu \int_\Omega \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v})\, dx + \lambda \int_\Omega \operatorname{div} \underline{u}\ \operatorname{div} \underline{v}\, dx \\ = \int_\Omega \underline{f} \cdot \underline{v}\, dx + \int_{\Omega_1} g_n \underline{v} \cdot \underline{n}\, ds + \int_{\Gamma_2} g_t \underline{v} \cdot \underline{t}\, ds, \quad \forall \underline{v} \in V, \end{aligned} \tag{1.2.32}$$

which is still clearly of the form (1.1.6). We now use the classical integration by parts formula

$$\int_\Omega \underline{\underline{m}} : \underline{\underline{\varepsilon}}(\underline{v})\, dx = -\int_\Omega (\operatorname{div} \underline{\underline{m}}) \cdot \underline{v}\, dx + \int_\Gamma m_{nn} \underline{v} \cdot \underline{n} ds + \int_\Gamma m_{nt} \underline{v} \cdot \underline{t}\, ds \tag{1.2.33}$$

where $\underline{\underline{m}}$ is a (smooth enough) tensor, and m_{nn} and m_{nt} denote the normal and tangential parts of the traction vector m_n, i.e.

$$\begin{cases} m_{nn} := \displaystyle\sum_{i,j} m_{ij}\, n_i\, n_j = \sum_i \Big\{ \sum_j m_{ij} n_j \Big\} n_i = \sum_i (\underline{m}_n)_i\, n_i, \\ m_{nt} := \displaystyle\sum_{i,j} m_{ij} t_i\, n_j = \sum_i \Big\{ \sum_j m_{ij} n_j \Big\} t_i = \sum_i (\underline{m}_n)_i\, t_i. \end{cases} \tag{1.2.34}$$

Equation (1.2.32) can now be interpreted as

$$\begin{cases} -(2\mu \operatorname{div} \underline{\underline{\varepsilon}}(\underline{u}) + \lambda \operatorname{grad} \operatorname{div} \underline{u}) = \underline{f} \text{ in } \Omega, \\ \underline{u}\big|_{\Gamma_0} = 0, \\ 2\mu \varepsilon_{nn} + \lambda\ \operatorname{div} \underline{u} = g_n, \\ 2\mu \varepsilon_{nt} = g_t \text{ on } \Gamma_1. \end{cases} \tag{1.2.35}$$

Let us now introduce the stress tensor $\underline{\underline{\sigma}} := \underline{\underline{s}}^D + p\underline{\underline{\delta}}$ and the constitutive law

$$\begin{cases} \underline{\underline{s}}^D := 2\mu \underline{\underline{\varepsilon}}^D(\underline{u}), \\ p := 2(\lambda + \mu)\ \operatorname{div} \underline{u}, \end{cases} \tag{1.2.36}$$

relating stresses to displacements. It is now clear that the first equation of (1.2.35) expresses the equilibrium condition of continuum mechanics,

$$\operatorname{div} \underline{\underline{\sigma}} + \underline{f} = 0. \tag{1.2.37}$$

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In applications, the constitutive law (1.2.36) will vary depending on the type of materials and will sometimes take very non linear forms. In this case, the expression of the energy functional (1.2.31) will change accordingly. Moreover, large displacements will require a much more complex treatment. Nevertheless, the problem described above remains valuable as a model for more complicated situations.

The case of an incompressible material is especially important. It leads to the same equations as in the study of viscous incompressible flows. □

Example 1.2.3 (Stokes' problem for viscous incompressible flow.). We now consider a low velocity flow of a viscous incompressible fluid in a domain Ω. We denote by $\underline{u}$ the velocity field and by $\underline{\underline{\varepsilon}}(\underline{u})$ the (linearised) strain rate tensor defined in the same way as in (1.2.27). We thus consider the minimisation problem with the same notation and the same space V as in Example 1.2.2, but now with the incompressibility condition $\operatorname{div} \underline{v} = 0$, that is,

$$\inf_{\substack{\underline{v} \in V \\ \operatorname{div} \underline{v} = 0}} \mu \int_\Omega |\underline{\underline{\varepsilon}}(\underline{v})|^2 - \int_\Omega \underline{f} \cdot \underline{v}\, dx + \int_{\Gamma_1} g_n\, \underline{v} \cdot \underline{n}\, ds + \int_{\Gamma_1} g_t\, \underline{v} \cdot \underline{t}\, ds. \quad (1.2.38)$$

As we shall see later, problem (1.2.31) can be considered, when λ is large, as an approximation (by a "penalty method") of problem (1.2.38). Indeed when λ is large the second constitutive relation of (1.2.36) forces, in some sense, $\operatorname{div} \underline{v}$ to be zero. In the limit for $\lambda = +\infty$ (1.2.36) becomes meaningless: we shall see in Sect. 1.3 that pressure can then be introduced as a Lagrange multiplier associated with the constraint $\operatorname{div} \underline{u} = 0$. □

We now present a fourth-order problem. It is again, from the physical point of view, an elasticity problem but in a special modelling.

Example 1.2.4 (Deflection of a thin clamped plate). We consider here the problem of a thin clamped plate deflected under a distributed load f. The physical model will be described in Chap. 10. We also refer to [147, 148] and [149] for more details on plate problems. Under reasonable assumptions (and setting, for simplicity, some physical constants equal to 1), one obtains that the vertical deflection ψ is solution of the minimisation problem

$$\inf_{\varphi \in H_0^2(\Omega)} \frac{1}{2} \int_\Omega |\Delta \varphi|^2\, dx - \int_\Omega f \varphi\, dx. \quad (1.2.39)$$

The unique solution ψ is characterised by

$$\int_\Omega \Delta \psi\, \Delta \varphi\, dx = \int_\Omega f \varphi\, dx, \quad \forall \varphi \in H_0^2(\Omega), \quad (1.2.40)$$

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and is the solution of the boundary value problem

$$\begin{cases} \Delta^2 \psi = f, \\ \psi|_\Gamma = 0, \\ \dfrac{\partial \psi}{\partial n}|_\Gamma = 0. \end{cases} \tag{1.2.41}$$

For these boundary conditions (representing a clamped plate) (1.2.39) is equivalent to

$$\inf_{\varphi \in H_0^2(\Omega)} \left\{ \frac{1}{2} \int_\Omega \left[\left\{ \frac{\partial^2 \varphi}{\partial x_1^2} \right\}^2 + 2 \left\{ \frac{\partial^2 \varphi}{\partial x_1 \partial x_2} \right\}^2 + \left\{ \frac{\partial^2 v \varphi}{\partial x_2^2} \right\}^2 \right] dx - \int_\Omega f \varphi \, dx \right\}, \tag{1.2.42}$$

which is, in general, more physically sound. These two mathematically equivalent forms can lead to different numerical methods. It must also be noted that *natural boundary conditions* (those arising from integration by parts) will not be the same if (1.2.39) and (1.2.42) are minimised on a space larger than $H_0^2(\Omega)$, so that the equivalence only holds for plates that are *clamped* all over the boundary. Actually the true potential energy of the plate (that is, the true functional which has to be minimised) is given, for a clamped plate, by

$$J(\varphi) := \frac{Et^3}{12(1-\nu^2)} \int_\Omega \Big\{ \nu |\Delta \varphi|^2 + (1-\nu) \Big[\Big(\frac{\partial^2 \varphi}{\partial x_1^2} \Big)^2 + 2 \Big(\frac{\partial^2 \varphi}{\partial x_1 \partial x_2} \Big)^2 + \Big(\frac{\partial^2 \varphi}{\partial x_2^2} \Big)^2 \Big] \Big\} \, dx - \int_\Omega f \varphi \, dx, \tag{1.2.43}$$

where E is Young's modulus, ν is the Poisson's coefficient and t is the thickness of the plate. In particular E and ν can be expressed in terms of the Lamé coefficients λ, μ in the following way

$$E := \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu}, \qquad \nu := \frac{\lambda}{2(\lambda + \mu)}. \tag{1.2.44}$$

We also recall that the Stokes problem (1.2.38) can also be expressed as a biharmonic problem by the introduction of a stream function ψ such that

$$\underline{u} = \Big\{ \frac{\partial \psi}{\partial x_2}, -\frac{\partial \psi}{\partial x_1} \Big\}. \tag{1.2.45}$$

We shall come back to this point in Sect. 1.3. □

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Example 1.2.5 (The time-harmonic Maxwell system). Maxwell's equations represent one of the most elegant and concise ways to state the fundamentals of electricity and magnetism. The classical electromagnetic field is described by the four vectors $\mathcal{E}$, $\mathcal{D}$, $\mathcal{H}$, and $\mathcal{B}$ which are functions of the position $\underline{x} \in \mathbb{R}^3$ and of the time $t \in \mathbb{R}$. The vectors $\mathcal{E}$ and $\mathcal{H}$ are referred to as the electric and magnetic field, while $\mathcal{D}$ and $\mathcal{B}$ are the electric and magnetic displacements, respectively.

The Faraday law of induction states

$$\int_{\partial\Sigma} \mathcal{E} \cdot ds = -\frac{d}{dt} \int_{\Sigma} \mathcal{B} \cdot \underline{n}, \tag{1.2.46}$$

for any closed orientable surface Σ in $\mathbb{R}^2$ with normal $\underline{n}$; namely, the circulation of the electric field equals the negative of the rate of change of the magnetic flux.

The Ampère law says

$$\int_{\partial\Sigma} \mathcal{H} \cdot ds = \frac{d}{dt} \int_{\Sigma} \mathcal{D} \cdot \underline{n} + \int_{\Sigma} \mathcal{J} \cdot \underline{n}, \tag{1.2.47}$$

with the same notation as above and where $\mathcal{J}$ denotes the current density vector.

From Eqs. (1.2.46) and (1.2.47) it can be noticed that the fields $\mathcal{E}$, $\mathcal{H}$, $\mathcal{B}$ and $\mathcal{D}$ have a different nature. Indeed, the first two are integral 1-forms, while the latter two are integral 2-forms in the spirit of [247] (Definition 1). This remark is of fundamental importance for the design of finite element schemes.

The differential forms of (1.2.46) and (1.2.47) read

$$\begin{aligned} &\frac{\partial \mathcal{B}}{\partial t} + \underline{\mathrm{curl}}\, \mathcal{E} = 0, \\ &\frac{\partial \mathcal{D}}{\partial t} - \underline{\mathrm{curl}}\, \mathcal{H} = -\mathcal{J}, \end{aligned} \tag{1.2.48}$$

which are usually referred to as Maxwell's equations, together with the two Gauss Laws

$$\begin{aligned} &\mathrm{div}\, \mathcal{D} = \rho, \\ &\mathrm{div}\, \mathcal{B} = 0, \end{aligned} \tag{1.2.49}$$

where ρ denotes the charge density function. It is clear that in (1.2.48) and (1.2.49) the quantities $\mathcal{J}$ and ρ cannot be taken independently: taking the divergence of the second equation in (1.2.48) and comparing with the time derivative of the first equation in (1.2.49) we have indeed:

$$\frac{\partial \rho}{\partial t} + \mathrm{div}\, \mathcal{J} = 0, \tag{1.2.50}$$

which could be seen as a *compatibility condition* when $\mathcal{J}$ and ρ are considered as *given data*.

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The time-harmonic Maxwell system is considered, for instance, when the Fourier transform in time is used or when the propagation of electromagnetic waves at a given frequency is studied. Then, given a frequency ω, we consider the *ansatz*:

$$\begin{aligned}
\mathcal{E}(\underline{x},t) &= \Re\left(e^{-i\omega t}E(\underline{x})\right),\\
\mathcal{D}(\underline{x},t) &= \Re\left(e^{-i\omega t}D(\underline{x})\right),\\
\mathcal{H}(\underline{x},t) &= \Re\left(e^{-i\omega t}H(\underline{x})\right),\\
\mathcal{B}(\underline{x},t) &= \Re\left(e^{-i\omega t}B(\underline{x})\right),
\end{aligned} \tag{1.2.51}$$

where $\Re$ denotes the *real part*. We define also

$$\begin{aligned}
\mathcal{J}(\underline{x},t) &= \Re\left(e^{-i\omega t}J(\underline{x})\right),\\
\rho(\underline{x},t) &= \Re\left(e^{-i\omega t}r(\underline{x})\right).
\end{aligned} \tag{1.2.52}$$

Standard constitutive equations for linear media read

$$D = \underline{\underline{\varepsilon}}E, \qquad B = \underline{\underline{\mu}}H, \tag{1.2.53}$$

where $\underline{\underline{\varepsilon}}$ and $\underline{\underline{\mu}}$ denote the electric permittivity and the magnetic permeability, respectively. For general inhomogeneous, anisotropic materials $\underline{\underline{\varepsilon}}$ and $\underline{\underline{\mu}}$ are 3×3 positive definite matrix functions.

Inserting constitutive relations (1.2.53) into (1.2.48) and (1.2.49), and considering the time-harmonic assumptions (1.2.51) and (1.2.52), we get the time harmonic Maxwell equations

$$\begin{aligned}
&\underline{\mathrm{curl}}\, E - i\omega\underline{\underline{\mu}}H = 0,\\
&\mathrm{div}(\underline{\underline{\varepsilon}}E) = r,\\
&\underline{\mathrm{curl}}\, H + i\omega\underline{\underline{\varepsilon}}E = J,\\
&\mathrm{div}(\underline{\underline{\mu}}H) = 0.
\end{aligned} \tag{1.2.54}$$

It is a standard procedure to eliminate one variable and to write (1.2.54) as a second order system. Eliminating for instance the field H, we get

$$\underline{\mathrm{curl}}(\underline{\underline{\mu}}^{-1}\,\underline{\mathrm{curl}}\, E) - \omega^2\underline{\underline{\varepsilon}}E = F, \tag{1.2.55}$$

where F is given by $i\omega J$, together with the divergence condition (which follows from the equation)

$$-\omega^2\,\mathrm{div}(\underline{\underline{\varepsilon}}E) = \mathrm{div}\, F. \tag{1.2.56}$$

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Equation (1.2.55) is usually equipped with suitable boundary conditions. The simplest one is the perfect conducting boundary condition which reads:

$$E \times \underline{n} = 0, \tag{1.2.57}$$

where $\underline{n}$ is the outward unit vector. We shall discuss variational formulations of the time harmonic Maxwell equations in Chap. 11. □

The examples presented above are among the most fundamental of mathematical physics and engineering problems. A good understanding of their properties will enable to extend the results obtained to more complex situations.

1.2.1 Eigenvalue Problems

The above examples can also be associated to eigenvalue problems. It is worth making them explicit as we shall be concerned later with alternate formulations and their numerical properties. In the case of Example 1.2.1, restricting ourselves to the case of Dirichlet's conditions, we have

$$\begin{cases} -\Delta p = \lambda p \text{ in } \Omega, \\ p\big|_{\Gamma} = 0. \end{cases} \tag{1.2.58}$$

The solutions of this problem describe, for instance, the vibrational modes of a membrane. It can be written in a more precise way as,

$$\int_{\Omega} \underline{\text{grad}}\, p \cdot \underline{\text{grad}}\, q \, dx = \lambda \int_{\Omega} p\, q \, dx, \quad \forall q \in H_0^1(\Omega). \tag{1.2.59}$$

It is classical [172, 382] that this problem has an infinite countable set of solutions $(p_k, \lambda_k, k \in N)$ with $\lambda_k \to \infty$ as k increases. The key to this result is the compact inclusion of $H_0^1(\Omega)$ into $L^2(\Omega)$.

In the same way, the eigenvalue problem associated with the elasticity problem of Example 1.2.2 is fundamental for the study of vibrations in elastic structures. The problem is then, restricting ourselves again to Dirichlet's conditions,

$$\begin{cases} -(2\mu \, \underline{\text{div}}\, \underline{\underline{\varepsilon}}(\underline{u}) + \lambda \, \underline{\text{grad}}\, \text{div}\, \underline{u}) = \tilde{\lambda} \underline{u} \text{ in } \Omega, \\ \underline{u}\big|_{\Gamma_0} = 0. \end{cases} \tag{1.2.60}$$

We have denoted the eigenvalue as $\tilde{\lambda}$ to distinguish it from the Lamé coefficients. The variational form is then,

$$2\mu \int_{\Omega} \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v}) \, dx + \lambda \int_{\Omega} \text{div}\, \underline{u} \ \text{div}\, \underline{v} \, dx = \tilde{\lambda} \int_{\Omega} \underline{u} \cdot \underline{v} \, dx \, ds, \qquad \forall \underline{v} \in V. \tag{1.2.61}$$

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Here again we have an infinite set of eigenvalues. Introducing in this problem the constraint $\operatorname{div} \underline{u} = 0$, we would get an eigenvalue problem for the Stokes problem of Example 1.2.3, or to a problem of incompressible elasticity as we shall see later. Eigenvalue problems related to Maxwell's equations will be discussed in Chap. 11.

1.3 Duality Methods

1.3.1 Generalities

Up to now, we have introduced equations that can be written as minimisation problems of some functionals in properly chosen functional spaces. This is the most classical way of setting these problems. Finite element approximations, based on the formulations described above, are routinely used in commercial codes. Various reasons justified the introduction, for these same problems and many other ones, of different variational formulations and therefore different finite element approximations. This was done at the beginning by many engineers. The reader may refer, for example to [321, 335, 336].

The first reason can be the presence in the variational formulation of a constraint, such as the condition $\operatorname{div} \underline{u} = 0$ in problem (1.2.38). As we shall see, it is difficult (and not necessary) to build finite element approximations satisfying exactly this constraint. It will be more efficient to modify the variational formulation and to introduce pressure.

A second reason may lie in the physical "importance" of the variables appearing in the problem. In elasticity problems, for example, it is often more useful to compute accurately stresses rather than displacements. In the standard formulation, stresses can be recovered from the displacements by (1.2.36) or some other similar law. Their computation requires the derivatives of the displacement field $\underline{u}$. From a numerical point of view, differentiating implies a loss of precision. It is therefore appealing to look for a formulation in which constraints are readily accessible.

A third reason comes from the difficulties arising in the discretisation of spaces of regular functions such as $H_0^2(\Omega)$ appearing in Example 1.2.4. Approximating this space by a finite element method implies ensuring continuity of the derivatives at interfaces between elements. This is possible but more cumbersome than approximating, say, $H^1(\Omega)$ or $H_0^1(\Omega)$. A variational formulation enabling to decompose a fourth-order problem into a system of second order problems permits to avoid building complicated elements, at the price of introducing some other difficulties.

Finally, a last reason could be to look for *a weaker variational formulation* corresponding better in some cases to available data (e.g. punctual loads) for which standard formulations may become meaningless due to a lack of regularity of the solution.

We must also point out that the "non-standard" formulations which we shall now describe have been initially introduced by engineers for one or some of the reasons discussed above. We quote in this respect, but in a totally non exhaustive way, [210, 243, 245, 320, 370]. On the other hand, very powerful tools for the transformation of

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variational problems can be found in convex analysis and duality theory [38,50,184, 338]. It is neither possible nor desirable to develop here duality theory and we shall restrict ourselves to the most basic facts. The fundamental idea of duality theory is that one can represent a convex function by the family of its tangent affine functions. This is indeed the principle of the classical Legendre transformation. More precisely, let us define for a given convex function $G(v)$, defined from a space V to $\mathbb{R}$, the conjugate function $G^*(v^*)$ on the dual space V' of V by

$$G^*(v^*) := \sup_{v\in V} \langle v, v^*\rangle_{V\times V'} - G(v). \tag{1.3.1}$$

Note that when $V = \mathbb{R}$, $G^*(v^*)$ is the intercept with the v axis of the tangent to G of slope v^*. The important point for what follows is that one can build $G(v)$ from $G^*(v^*)$ by the following formula, symmetrical to (1.3.1)

$$G(v) = \sup_{v^*\in V'} \langle v^*, v\rangle_{V'\times V} - G^*(v^*). \tag{1.3.2}$$

Given then a problem of the form

$$\inf_{v\in V} F(v) + G(v), \tag{1.3.3}$$

we can use (1.3.2) to obtain

$$\inf_{v\in V}\Big\{F(v) + \sup_{v^*\in V'} \langle v^*, v\rangle_{V'\times V} - G^*(v^*)\Big\} \tag{1.3.4}$$

that is, the saddle point problem

$$\inf_{v\in V}\sup_{v^*\in V'} F(v) + \langle v^*, v\rangle_{V'\times V} - G^*(v^*). \tag{1.3.5}$$

Under simple regularity assumptions, one can also consider the *dual problem*

$$\sup_{v^*\in V'}\Big\{\inf_{v\in V} F(v) + \langle v^*, v\rangle_{V'\times V} - G^*(v^*)\Big\}. \tag{1.3.6}$$

To fix ideas, it is worth considering a special and important case where we have as in (1.1.2), for $f \in V'$,

$$F(v) := \frac{1}{2}a(u, v) - \langle f, v\rangle. \tag{1.3.7}$$

We then introduce another Hilbert space Q and an operator B from V into Q' defined by a continuous bilinear form on $V \times Q$,

$$\langle Bv, q\rangle = b(v, q). \tag{1.3.8}$$

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We then want to solve for $g \in Q'$ the constrained problem

$$\inf_{Bv=g} J(v). \tag{1.3.9}$$

This constrained problem can be written as an unconstrained problem, introducing the characteristic function $\delta(\cdot|\{0\})$ defined on Q' by

$$\delta(q|\{g\}) := \begin{cases} 0 & \text{if } v = g, \\ +\infty & \text{otherwise.} \end{cases} \tag{1.3.10}$$

We can then write (1.3.9) as

$$\inf_{v\in V} F(v) + \delta(Bv|\{g\}), \tag{1.3.11}$$

which can be readily transformed into the saddle-point problem

$$\inf_{v\in V}\sup_{q\in Q} \frac{1}{2}a(u,v) - b(v,q) - \langle f, v\rangle_{V'\times V} + \langle g, q\rangle_{Q'\times Q}, \tag{1.3.12}$$

for which the optimality system is

$$\begin{cases} a(u,v) + b(v,p) = \langle f, v\rangle_{V'\times V}, & \forall v \in V, \\ b(u,q) = \langle g, q\rangle_{Q'\times Q}, & \forall q \in Q, \end{cases} \tag{1.3.13}$$

or in operator form, denoting A the operator from V into V' defined by $a(\cdot,\cdot)$,

$$\begin{cases} Au + B^t p = f, \\ Bu = g. \end{cases} \tag{1.3.14}$$

Remark 1.3.1. Problem (1.3.12) has the general form

$$\inf_{v\in V}\sup_{q\in Q} L(v,q), \tag{1.3.15}$$

where $L(v,q)$ is a convex-concave functional on $V \times Q$. If one first eliminates q by computing

$$J(v) = \sup_{q\in Q} L(v,q),$$

one falls back on the original problem, the *primal problem*. Reversing the order of operations, (this cannot always be done, but no problems arise in the examples we present) and eliminating v from $L(v,q)$ by defining

$$D(q) := \inf_{v} L(v,q) \tag{1.3.16}$$

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leads to the dual problem

$$\sup_{q\in Q} D(q). \tag{1.3.17}$$

□

The dual problem in Q corresponding to (1.3.12) would take the form

$$\inf_{q} \frac{1}{2}\langle A^{-1}B^t q, B^t q\rangle_{V\times V'} - \langle A^{-1}f, B^t q\rangle_{V\times V'} + \langle g, q\rangle_{Q'\times Q}, \tag{1.3.18}$$

or in operator form,

$$BA^{-1}B^t p = BA^{-1}f - g. \tag{1.3.19}$$

We now apply this idea to the previous examples. This form of problem and its variants will be central to our study and we shall proceed to introduce some examples.

1.3.2 Examples for Symmetric Problems

Example 1.3.1 (Introduction of pressure in Stokes' problem.). Let us consider problem (1.2.38) in which, to make the presentation easier, we take $\Gamma_0 = \Gamma$ that is pure Dirichlet conditions on the boundary. This constrained problem can be written as an unconstrained problem, introducing as above the characteristic function $\delta(\cdot|\{0\})$ defined on $L^2(\Omega)$ by

$$\delta(v|\{0\}) := \begin{cases} 0 & \text{if } v = 0, \\ +\infty & \text{otherwise.} \end{cases} \tag{1.3.20}$$

It is thus a pure change of notations to write instead of (1.2.38)

$$\inf_{v\in V} \mu \int_\Omega |\underline{\underline{\varepsilon}}(\underline{v})|^2\, dx - \int_\Omega \underline{f}\cdot\underline{v}\, dx + \delta(\operatorname{div}\underline{v}|\{0\}), \tag{1.3.21}$$

where $V = (H_0^1(\Omega))^2$. On the other hand, one clearly has,

$$\delta(\operatorname{div}\underline{u}|\{0\}) = \sup_{q\in L^2(\Omega)} \int_\Omega q \operatorname{div}\underline{u}\, dx, \quad \forall \underline{u}\in (H_0^1(\Omega))^2, \tag{1.3.22}$$

and the minimisation problem (1.3.21) can be transformed into the saddle point problem,

$$\inf_{v\in V}\sup_{q\in L^2(\Omega)} \mu \int_\Omega |\underline{\underline{\varepsilon}}(\underline{v})|^2\, dx - \int_\Omega \underline{f}\cdot\underline{v}\, dx - \int_\Omega q \operatorname{div}\underline{v}\, dx. \tag{1.3.23}$$

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This apparently simple trick has in reality completely changed the nature of the problem. We now have to find a pair $(\underline{u}, p)$ solution of the variational system

$$\begin{cases} 2\mu \int_\Omega \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v})\, dx - \int_\Omega \underline{f} \cdot \underline{v}\, dx - \int_\Omega p \ \mathrm{div}\ \underline{v}\, dx = 0, \quad \forall \underline{v} \in V, \\ \int_\Omega q \ \mathrm{div}\ \underline{u}\, dx = 0, \quad \forall q \in L^2(\Omega). \end{cases} \tag{1.3.24}$$

The second equation of (1.3.24) evidently expresses the condition $\mathrm{div}\ \underline{u} = 0$. In order to use (1.3.24), we shall have to show the existence of a saddle point $(\underline{u}, p)$, and in particular the existence of the Lagrange multiplier p. This will be done in Chap. 4. The variational system (1.3.24) can be interpreted in the form

$$\begin{cases} -2\mu A\underline{u} + \underline{\mathrm{grad}}\ p = \underline{f}, \\ \mathrm{div}\ \underline{u} = 0, \\ \underline{u}|_\Gamma = 0, \end{cases} \tag{1.3.25}$$

where we used the operator $A\underline{u} = \mathrm{div}\, \underline{\underline{\varepsilon}}(\underline{u})$

$$A\underline{u} = \begin{pmatrix} \dfrac{\partial^2 u_1}{\partial x_1^2} + \dfrac{\partial}{\partial x_2} \dfrac{1}{2} \left\{ \dfrac{\partial u_1}{\partial x_2} + \dfrac{\partial u_2}{\partial x_1} \right\} \\ \dfrac{\partial^2 u_2}{\partial x_2^2} + \dfrac{\partial}{\partial x_1} \dfrac{1}{2} \left\{ \dfrac{\partial u_1}{\partial x_2} + \dfrac{\partial u_2}{\partial x_1} \right\} \end{pmatrix}. \tag{1.3.26}$$

Under the *divergence-free condition* $\mathrm{div}\ \underline{u} = 0$, this can also be written as

$$\begin{cases} -\mu \Delta \underline{u} + \underline{\mathrm{grad}}\ p = \underline{f}, \\ \mathrm{div}\ \underline{u} = 0, \end{cases} \tag{1.3.27}$$

which is the classical form of the Stokes problem. □

Example 1.3.2 (Dual problem for the Stokes problem). In the case of the Stokes problem, the dual problem can be expressed, as we shall see, in many equivalent ways. In order to find it we must, q being given, find the minimum in respect to $\underline{v}$ of $L(\underline{v}, q) = \mu \int_\Omega |\underline{\underline{\varepsilon}}(\underline{v})|^2\, dx - \int_\Omega \underline{f} \cdot \underline{v}\, dx - \int_\Omega q\ \mathrm{div}\ \underline{v}\, dx$. The minimum, that we denote by $\underline{u}_q$, is characterised by

$$2\mu \int_\Omega \underline{\underline{\varepsilon}}(\underline{u}_q) : \underline{\underline{\varepsilon}}(\underline{v})\, dx - \int_\Omega \underline{f} \cdot \underline{v}\, dx - \int_\Omega q\, \mathrm{div}\, \underline{v}\, dx = 0, \quad \forall \underline{v} \in V. \tag{1.3.28}$$

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Taking $\underline{v} = \underline{u}_q$ this gives,

$$2\mu \int_\Omega |\underline{\underline{\varepsilon}}(\underline{u}_q)|^2 \, dx - \int_\Omega \underline{f} \cdot \underline{u}_q \, dx - \int_\Omega q \, \mathrm{div}\, \underline{u}_q \, dx = 0. \tag{1.3.29}$$

Using (1.3.29) to evaluate $L(\underline{u}_q, q)$, the dual problem can be written as an optimal control problem,

$$\sup_{q \in L^2(\Omega)} -\mu \int_\Omega |\underline{\underline{\varepsilon}}(\underline{u}_q)|^2 \, dx, \tag{1.3.30}$$

where $\underline{u}_q$ is the solution of

$$\begin{cases} -2\mu \, A\underline{u}_q + \underline{\mathrm{grad}}\, q = \underline{f}, \\ \underline{u}_q|_\Gamma = 0. \end{cases} \tag{1.3.31}$$

Denoting by G the Green operator defining the solution of (1.3.31), that is

$$\underline{u}_q = G(\underline{f} - \underline{\mathrm{grad}}\, q) \tag{1.3.32}$$

and using (1.3.32) in (1.3.30), one can get from (1.3.29)

$$\inf_q \int_\Omega \underline{\mathrm{grad}}\, q \cdot G(\underline{\mathrm{grad}}\, q) \, dx - \int_\Omega G(\underline{f}) \cdot \underline{\mathrm{grad}}\, q \, dx. \tag{1.3.33}$$

One notices that this dual problem is a problem in $\underline{\mathrm{grad}}\, q$. It is well-known that the solution p is defined (for Dirichlet conditions on $\underline{u}$) only up to an additive constant. One can interpret (1.3.33) as the equation,

$$\mathrm{div}(G\, \underline{\mathrm{grad}}\, q) = \mathrm{div}(G\, \underline{f}). \tag{1.3.34}$$

If one defines on $V' := (H^{-1}(\Omega))^2$ the norm

$$\|\underline{f}\|_G^2 := \langle G\underline{f}, \underline{f} \rangle_{V \times V'}, \tag{1.3.35}$$

problem (1.3.33) can be written as a least-squares problem

$$\inf_{q \in L^2(\Omega)} \frac{1}{2} \| \underline{\mathrm{grad}}\, q - \underline{f} \|_G^2. \tag{1.3.36}$$

□

The presence of a Green operator makes this dual problem difficult to handle directly. It is however implicitly the basis of some numerical solution procedures [205, 368]. We shall meet below other dual problems that will have a large direct importance and that will be handled as such.

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Example 1.3.3 (A duality method for nearly incompressible elasticity). We already noted in Examples 1.2.2 and 1.2.3 that the linear elasticity problem and the Stokes problem are very similar when a nearly incompressible material is considered. We now develop this analogy in the framework of Example 1.3.1. The starting point will be the obvious result,

$$\frac{\lambda}{2}\int_\Omega |\operatorname{div} \underline{v}|^2\,dx = \sup_{q\in L^2(\Omega)} \int_\Omega q \operatorname{div} \underline{v}\,dx - \frac{1}{2\lambda}\int_\Omega |q|^2\,dx. \tag{1.3.37}$$

Substituting (1.3.37) into (1.2.31) we get, by the same methods as in the previous examples, the problem

$$\begin{aligned}\inf_{\underline{v}\in V}\sup_{q\in L^2(\Omega)} \mu\int_\Omega |\underline{\underline{\varepsilon}}(\underline{v})|^2\,dx + \int_\Omega q\ \operatorname{div}\ \underline{v}\,dx - \frac{1}{2\lambda}\int_\Omega |q|^2\,dx \\ - \int_\Omega \underline{f}\cdot\underline{v}\,dx - \int_{\Gamma_1} g_n\,\underline{v}\cdot\underline{n}\,ds - \int_{\Gamma_1} g_t\,\underline{v}\cdot\underline{t}\,ds.\end{aligned} \tag{1.3.38}$$

The solution $(\underline{u}, p)$ of problem (1.3.38) is characterised by the system,

$$\begin{cases} 2\mu\displaystyle\int_\Omega \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v})\,dx + \int_\Omega p \operatorname{div}\underline{v}\,dx \\ \qquad = \displaystyle\int_\Omega \underline{f}\cdot\underline{v}\,dx + \int_{\Gamma_1}(g_n\underline{v}\cdot\underline{n} + g_t\underline{v}\cdot\underline{t})\,ds, \quad \forall \underline{v}\in V, \\ \displaystyle\int_\Omega q\operatorname{div}\underline{u}\,dx - \frac{1}{\lambda}\int_\Omega pq\ dx = 0, \quad \forall q\in L^2(\Omega). \end{cases} \tag{1.3.39}$$

This can be summarised by saying that we transformed our original problem into a system by introducing the auxiliary variable $p = \lambda\ \operatorname{div}\ \underline{u}$. It must be noted that this also makes our minimisation problem become a saddle point problem. We shall see in Chap. 8, that this apparently *tautological change has implications in the building of numerical approximations to* (1.2.31) *that remain valid when* λ *is large.* □

Example 1.3.4 (Dualisation of the Poisson problem). The result that we shall get here can be obtained by many methods. Techniques of convex analysis permit one to extend what appears to be a trick to much more complex situations. However it will be sufficient for our purpose to follow the simple development below. Let us then consider the Dirichlet problem,

$$\inf_{q\in H_0^1(\Omega)} \frac{1}{2}\int_\Omega |\operatorname{\underline{grad}} q|^2\,dx - \int_\Omega fq\,dx. \tag{1.3.40}$$

In many applications $\operatorname{\underline{grad}} p$ rather than p is the interesting variable. For instance in thermo-diffusion problems, $\operatorname{\underline{grad}} p$ will be the heat flux which is (very often) more important to know than the temperature p. What we now do is essentially to

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introduce the auxiliary variable $\underline{u} = \underline{\text{grad}}\, p$ to transform our problem into a system. To do so, we use the same trick as in Example 1.3.3 and write

$$\frac{1}{2}\int_\Omega |\ \underline{\text{grad}}\, q|^2\, dx = \sup_{\underline{v}\in(L^2(\Omega))^2} \int_\Omega \underline{v}\cdot\ \underline{\text{grad}}\, q\, dx - \frac{1}{2}\int_\Omega |\underline{v}|^2\, dx, \tag{1.3.41}$$

which we use in (1.3.40) to get the saddle point problem,

$$\inf_{q\in Q}\sup_{\underline{v}\in V} -\frac{1}{2}\int_\Omega |\underline{v}|^2\, dx - \int_\Omega fq\, dx + \int_\Omega \underline{v}\cdot\ \underline{\text{grad}}\, q\, dx, \tag{1.3.42}$$

where $Q := H_0^1(\Omega)$ and $V := (L^2(\Omega))^2$. The saddle point $(p, \underline{u})$ is characterised by

$$\begin{cases} \displaystyle\int_\Omega \underline{u}\cdot\underline{v}\, dx - \int_\Omega \underline{v}\cdot\ \underline{\text{grad}}\, p\, dx = 0 \quad \forall \underline{v}\in V, \\ \displaystyle\int_\Omega \underline{u}\cdot\ \underline{\text{grad}}\, q\, dx = \int_\Omega fq\, dx, \quad \forall q\in Q, \end{cases} \tag{1.3.43}$$

and this can be read as

$$\begin{cases} \underline{u} = \ \underline{\text{grad}}\, p,\ p\in H_0^1(\Omega), \\ \text{div}\ \underline{u} + f\ = 0, \end{cases} \tag{1.3.44}$$

which is evidently equivalent to a standard Dirichlet problem for Laplace operator.

The *dual problem* is readily made explicit. Writing it as a minimisation problem by changing the sign of the objective functional, we have

$$\inf_{\underline{v}\in Z_f} \frac{1}{2}\int_\Omega |\underline{v}|^2\, dx, \quad \text{where}\ \ Z_f := \{\underline{v}\in (L^2(\Omega))^2|\ \text{div}\ \underline{v} + f\ = 0\}. \tag{1.3.45}$$

This is the classical *complementary energy principle.* □

We now want to get a weaker form of this problem. In order to do so, we recall a functional space already introduced in (1.2.16):

$$H(\text{div};\Omega) := \{\underline{v}|\ \underline{v}\in (L^2(\Omega))^2,\ \text{div}\ \underline{v}\in L^2(\Omega)\}, \tag{1.3.46}$$

and we consider on it the following norm

$$\|\underline{v}\|^2_{H(\text{div};\Omega)} := \|\underline{v}\|^2_{0,\Omega} + \|\,\text{div}\ \underline{v}\|^2_{0,\Omega}, \tag{1.3.47}$$

that makes it a Hilbert space. As we have already said, vectors of $H(\text{div};\Omega)$ admit a well defined *normal trace on* $\Gamma := \partial\Omega$. This normal trace $\underline{v}\cdot\underline{n}$, lies in $H^{-1/2}(\Gamma)$ and one has the following "integration by parts" formula,

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$$\int_\Omega \underline{v} \cdot \underline{\mathrm{grad}}\, q \, dx + \int_\Omega \mathrm{div}\, \underline{v}\, q \, dx = \langle q, \underline{v} \cdot \underline{n} \rangle_{H^{\frac{1}{2}}(\Gamma) \times H^{-\frac{1}{2}}(\Gamma)}, \tag{1.3.48}$$

for any $\underline{v} \in H(\mathrm{div}; \Omega)$ and any $q \in H^1(\Omega)$. We shall often write formally $\int_\Gamma q \underline{v} \cdot \underline{n}\, ds$ instead of the duality product $\langle q,\ \underline{v} \cdot \underline{n} \rangle$.

Remark 1.3.2. The norm (1.3.47) is surely as *weird* as many other Sobolev norms. See Remark 1.2.1. The *proper way* to write it would be to take a characteristic length ℓ of the problem (for instance, the diameter of Ω), and consider instead

$$\|\underline{v}\|^2_{H(\mathrm{div};\Omega)} := \|\underline{v}\|^2_{0,\Omega} + \ell^2 \| \mathrm{div}\, \underline{v}\|^2_{0,\Omega}. \tag{1.3.49}$$

We do not do it here, nor, in general, throughout the book, in order to have simpler formulae to deal with. But, nevertheless, we consider it as *not healthy*. □

Example 1.3.5 (Weak form of the dual Poisson problem). If we take $f \in L^2(\Omega)$, problem (1.3.45) which is a constrained problem can be changed into a saddle point problem, as in Example 1.3.1, by introducing a Lagrange multiplier $p \in L^2(\Omega)$, that is, as $\underline{v}$ now belongs to $H(\mathrm{div}; \Omega)$,

$$\inf_{\underline{v} \in H(\mathrm{div};\Omega)} \sup_{q \in L^2(\Omega)} \frac{1}{2} \int_\Omega |\underline{v}|^2 \, dx + \int_\Omega fq \, dx + \int_\Omega q \ \mathrm{div}\, \underline{v} \, dx. \tag{1.3.50}$$

The functional spaces employed precisely enable us to write every term in (1.3.50) without ambiguity. We now look for a saddle point $(p, \underline{u})$ satisfying the variational system,

$$\begin{cases} \displaystyle\int_\Omega \underline{u} \cdot \underline{v} \, dx + \int_\Omega p \ \mathrm{div}\, \underline{v} \, dx = 0 \quad \forall \underline{v} \in H(\mathrm{div}; \Omega), \\ \displaystyle\int_\Omega (\mathrm{div}\, \underline{u} + f) q \, dx = 0 \quad \forall q \in L^2(\Omega). \end{cases} \tag{1.3.51}$$

Using (1.3.48) with $\underline{v} \cdot \underline{n}|_\Gamma = 0$, we obtain from (1.3.51)

$$\underline{u} = \underline{\mathrm{grad}}\, p. \tag{1.3.52}$$

Now $p \in L^2(\Omega)$ and $\underline{\mathrm{grad}}\, p = \underline{u} \in (L^2(\Omega))^2$ imply that $p \in H^1(\Omega)$ and it is justified to consider its trace. Using again (1.3.48) with a general $\underline{v}$ shows that $p|_\Gamma = 0$. The solution of our "weaker" problem is then the solution of the standard problem. However the discretisations of problem (1.3.51) will be quite different from those used for the standard formulation. □

Remark 1.3.3. The previous formulation enables us to write directly in a variational form a non-homogeneous Dirichlet problem. Indeed the solution $(p, \underline{u})$ of the saddle point problem with $g \in H^{\frac{1}{2}}(\Gamma)$,

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$$\inf_{\underline{v}} \sup_{q} \frac{1}{2} \int_{\Omega} |\underline{v}|^2 \, dx + \int_{\Omega} (\operatorname{div} \underline{v} + f) q \, dx - \int_{\Gamma} g \underline{v} \cdot \underline{n} \, ds, \tag{1.3.53}$$

leads to $\underline{u} = \operatorname{grad} p$, $\operatorname{div} \underline{u} + f = 0$, $p|_{\Gamma} = g$.

On the other hand, Neumann conditions become *essential* conditions that have to be incorporated into the construction of $\underline{u}$, that is in the choice of the functional space. □

Example 1.3.6 (Dualisation for the linear elasticity problem). We now want to extend the previous results to the case of the *linear elasticity problem*. We shall thus get a second way to dualise problem (1.2.31). It is a general fact that there is no unique way to use duality techniques.

The lines of the development are the same as for the Poisson problem and we shall avoid to write the details. We just point out that now $\underline{u}$ and $\underline{v}$ play the role of p and q, while $\underline{\underline{\sigma}}$ and $\underline{\underline{\tau}}$ play the role of $\underline{u}$ and $\underline{v}$, respectively.

We start by introducing the space

$$\underline{\underline{H}}(\underline{\operatorname{div}}; \Omega)_s := \{\underline{\underline{\tau}} \mid \tau_{ij} \in L^2(\Omega), \ \tau_{ij} = \tau_{ji}, \ \underline{\operatorname{div}} \underline{\underline{\tau}} \in (L^2(\Omega))^2\}, \tag{1.3.54}$$

where $\underline{\operatorname{div}} \underline{\underline{\tau}}$ is the *vector* $\frac{\partial}{\partial x_1} \tau_{i1} + \frac{\partial}{\partial x_2} \tau_{i2}$. On this space we use the norm,

$$\|\underline{\underline{\tau}}\|^2_{\underline{\underline{H}}(\underline{\operatorname{div}}; \Omega)_s} := \sum_{i,j} \int_{\Omega} |\tau_{ij}|^2 \, dx + \|\underline{\operatorname{div}} \underline{\underline{\tau}}\|^2_{(L^2(\Omega))^2}, \tag{1.3.55}$$

which makes it a Hilbert space.

One can then define, as for $H(\operatorname{div}; \Omega)$, the vector $\underline{\tau}_n \in (H^{-\frac{1}{2}}(\Gamma))^2$

$$(\underline{\tau}_n)_i := \sum_j \tau_{ij} \, n_j \tag{1.3.56}$$

and we shall mostly use the normal and tangential components, τ_{nn} and τ_{nt}, of this vector, as defined in (1.2.34). We then have the following "integration by parts" formula,

$$\int_{\Omega} \underline{\underline{\tau}} : \underline{\underline{\varepsilon}}(\underline{v}) \, dx + \int_{\Omega} \underline{\operatorname{div}} \underline{\underline{\tau}} \cdot \underline{v} \, dx = \langle \underline{\tau}_n, \underline{v} \rangle = \langle \tau_{nn}, \underline{v} \cdot \underline{n} \rangle + \langle \tau_{nt}, \underline{v} \cdot \underline{t} \rangle, \tag{1.3.57}$$

which is valid for any $\underline{\underline{\tau}}$ and $\underline{v}$ smooth enough. We have denoted $\langle \cdot, \cdot \rangle$ the duality between $H^{-\frac{1}{2}}(\Gamma)$ and $H^{\frac{1}{2}}(\Gamma)$ and shall often write the formal expression $\int_{\Gamma} \tau_{nn} \, \underline{v} \cdot \underline{n} \, ds + \int_{\Gamma} \tau_{nt} \, \underline{v} \cdot \underline{t} \, ds$.

We can now write our dual formulation for the linear elasticity problem. Following the same line as for the Poisson problem, we now write

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$$\mu \int_\Omega |\underline{\underline{\varepsilon}}(\underline{v})|^2\, dx + \frac{\lambda}{2} \int_\Omega |\operatorname{div} \underline{v}|^2\, dx = \sup_{\underline{\underline{\tau}} \in \underline{\underline{H}}(\operatorname{div};\Omega)_s} \int_\Omega \underline{\underline{\tau}}^D : \underline{\underline{\varepsilon}}^D\, dx + \int_\Omega tr\, \underline{\underline{\tau}}\, tr\, \underline{\underline{\varepsilon}}\, dx - \frac{1}{4\mu} \int_\Omega |\underline{\underline{\tau}}^D|^2\, dx - \frac{1}{2(\lambda+\mu)} \int_\Omega (tr\, \underline{\underline{\tau}})^2\, dx, \tag{1.3.58}$$

which leads us to the saddle point problem in $\underline{\underline{H}}(\operatorname{div};\Omega) \times (L^2(\Omega))^2$,

$$\inf_{\underline{\underline{\tau}}} \sup_{\underline{v}} \frac{1}{2(\lambda+\mu)} \int_\Omega |tr\, \underline{\underline{\tau}}|^2\, dx + \frac{1}{4\mu} \int_\Omega |\underline{\underline{\tau}}^D|^2\, dx + \int_\Omega (\operatorname{div} \underline{\underline{\tau}} + \underline{f})\, \underline{v}\, dx. \tag{1.3.59}$$

The solution $(\underline{\underline{\sigma}}, \underline{u})$ of this saddle point problem is characterised by the system

$$\begin{cases} \operatorname{div} \underline{\underline{\sigma}} + \underline{f} = 0, \\ tr\, \underline{\underline{\sigma}} = (\lambda+\mu)\, tr\, \underline{\underline{\varepsilon}}(\underline{u}), \\ \underline{\underline{\sigma}}^D = 2\mu\, \underline{\underline{\varepsilon}}^D(\underline{u}), \end{cases} \tag{1.3.60}$$

which are the equilibrium condition (1.2.37) and the constitutive relations (1.2.36). The dual problem then consists in minimising the *complementary energy*

$$\inf_{\underline{\underline{\tau}}} \frac{1}{4\mu} \int_\Omega |\underline{\underline{\tau}}^D|^2\, dx + \frac{1}{2(\lambda+\mu)} \int_\Omega |tr\, \underline{\underline{\tau}}|^2\, dx, \tag{1.3.61}$$

under the constraint $\operatorname{div} \underline{\underline{\tau}} + \underline{f} = 0$. Both the *mixed formulation* (1.3.59) and the dual formulation (1.3.61) are used in practice. They lead to different although similar approximations. □

We now consider the thin plate problem of Example 1.2.4 to introduce a mixed formulation due to [152] and [298].

Example 1.3.7 (Decomposition of a biharmonic problem). Again using the same technique as in the dualisation of the Dirichlet problem in Example 1.3.4, it is a simple exercise to transform problem (1.2.39) into the saddle point problem

$$\inf_{\mu \in L^2(\Omega)} \sup_{\varphi \in H_0^2(\Omega)} \frac{1}{2} \int_\Omega |\mu|^2\, dx + \int_\Omega \mu\, \Delta\varphi\, dx + \int_\Omega f\varphi\, dx, \tag{1.3.62}$$

and to get the dual problem,

$$\inf_{\mu \in M} \frac{1}{2} \int_\Omega |\mu|^2\, dx, \tag{1.3.63}$$

where $M := \{\mu \in L^2(\Omega),\ \Delta\mu + f = 0\}$. Integrating by parts the term $\int_\Omega \mu\, \Delta\varphi\, dx$, we get, as in Example 1.3.5, a weaker formulation

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$$\inf_{\mu\in L^2(\Omega)} \sup_{\varphi\in H_0^2(\Omega)} \frac{1}{2}\int_\Omega |\mu|^2\,dx - \int_\Omega \underline{\mathrm{grad}}\,\mu\cdot\,\underline{\mathrm{grad}}\,\varphi\,dx + \int_\Omega f\varphi\,dx. \quad (1.3.64)$$

Assume that (1.3.64) has a saddle point (ω,ψ) with $\omega\in H^1(\Omega)$. Then (ω,ψ) is characterised by the variational system

$$\begin{cases} \displaystyle\int_\Omega \omega\mu\,dx - \int_\Omega \underline{\mathrm{grad}}\,\mu\cdot\,\underline{\mathrm{grad}}\,\psi\,dx = 0 & \forall\mu\in H^1(\Omega),\\ \displaystyle\int_\Omega \underline{\mathrm{grad}}\,\omega\cdot\,\underline{\mathrm{grad}}\,\varphi\,dx = \int_\Omega f\varphi\,dx & \forall\varphi\in H_0^1(\Omega). \end{cases} \quad (1.3.65)$$

It is not difficult to see that the two equations of (1.3.65) imply

$$\begin{cases} -\Delta\psi = \omega & \text{and} \quad \dfrac{\partial\psi}{\partial n}\Big|_\Gamma = 0,\\ -\Delta\omega = f. \end{cases} \quad (1.3.66)$$

As we already have $\psi|_\Gamma = 0$ (since $\psi\in H_0^1(\Omega)$), we have in (1.3.66) too many boundary conditions on ψ and none on ω. The system however has a solution (ω,ψ) (provided Ω and f are smooth enough) such that the solution of the Dirichlet problem in ψ also satisfies (through the choice of the right-hand side) the extra Neumann condition. □

Example 1.3.8 (Decomposition of the plate bending problem). We now consider the plate bending problem (1.2.43). In order to make the dual problem easier to introduce, we first write the energy functional in the form

$$\frac{1}{2}\Big(\frac{Et^3}{12(1-\nu^2)}\Big)\int_\Omega \mathfrak{M}(\underline{\underline{D}}_2\varphi) : \underline{\underline{D}}_2\varphi\,dx - \int_\Omega f\varphi\,dx, \quad (1.3.67)$$

where the operator $\underline{\underline{D}}_2$ is defined by

$$(\underline{\underline{D}}_2\varphi)_{ij} := \frac{\partial^2\varphi}{\partial x_i\partial x_j},\ 1\le i,\ j\le 2, \quad (1.3.68)$$

and the operator $\mathfrak{M}$ by

$$\mathfrak{M}(\underline{\underline{\tau}}) := \begin{pmatrix} \tau_{11}+\nu\tau_{22} & (1-\nu)\tau_{12}\\ (1-\nu)\tau_{12} & \nu\tau_{11}+\tau_{22}\end{pmatrix}, \quad (1.3.69)$$

for any symmetric tensor $\underline{\underline{\tau}}$. Using the same kind of analysis as in the previous examples, we then get the saddle point problem

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$$\inf_{\underline{\underline{\tau}}\in(L^2(\Omega))^4_s} \sup_{\varphi\in H^2_0(\Omega)} \frac{1}{2}\Big(\frac{12(1-\nu^2)}{Et^3}\Big)\int_\Omega \mathfrak{M}^{-1}(\underline{\underline{\tau}}) : \underline{\underline{\tau}}\, dx + \int_\Omega \underline{\underline{\tau}} : \underline{\underline{D}}_2\varphi\, dx - \int_\Omega f\varphi\, dx, \quad (1.3.70)$$

where $(L^2(\Omega))^4_s$ is the space of square integrable 2×2 symmetric tensors. We introduce, as dual variables, the bending moments, obtained from the second derivatives of the primal solution ψ by

$$\underline{\underline{\sigma}} := -\frac{Et^3}{12(1-\nu^2)}\mathfrak{M}(\underline{\underline{D}}_2 u), \quad (1.3.71)$$

or explicitly

$$\begin{cases} \sigma_{11} = -\dfrac{Et^3}{12(1-\nu^2)}\Big(\dfrac{\partial^2\psi}{\partial x_1^2} + \nu\dfrac{\partial^2\psi}{\partial x_2^2}\Big), \\ \sigma_{22} = -\dfrac{Et^3}{12(1-\nu^2)}\Big(\nu\dfrac{\partial^2\psi}{\partial x_1^2} + \dfrac{\partial^2\psi}{\partial x_2^2}\Big), \\ \sigma_{12} = -\dfrac{Et^3}{12(1+\nu)}\dfrac{\partial^2\psi}{\partial x_1\partial x_2}. \end{cases} \quad (1.3.72)$$

The dual problem can then be written as

$$\inf_{\underline{\underline{\tau}}} \frac{1}{2}\Big(\frac{12}{Et^3}\Big)\int_\Omega [(\tau_{11}+\tau_{22})^2 + 2(1+\nu)(\tau_{12}^2 - \tau_{11}\tau_{22})]\, dx, \quad (1.3.73)$$

under the constraint

$$D_2^*\underline{\underline{\tau}} = f. \quad (1.3.74)$$

In (1.3.74) we denoted by D_2^* the transpose of the operator $\underline{\underline{D}}_2$ so that

$$D_2^*\underline{\underline{\tau}} = \frac{\partial^2\tau_{11}}{\partial x_1^2} + 2\frac{\partial^2\tau_{12}}{\partial x_1\partial x_2} + \frac{\partial^2\tau_{22}}{\partial x_2^2}. \quad (1.3.75)$$

It is possible, as in the previous case, to integrate by parts the expressions (1.3.74) and to obtain formulations in different functional spaces. We shall see an example of such a procedure in Sect. 10.2. □

1.3.3 Duality Methods for Non Symmetric Bilinear Forms

In all previous examples, our variational formulations were based on a minimisation problem for a functional and we were led to introduce a genuine saddle point

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problem. Even if this classical framework is suitable for a first presentation, it is not the sole possible, and the techniques developed before can also be applied to problems which are not optimisation problems. Let us consider for instance in $H_0^1(\Omega)$ a *continuous* and *coercive* bilinear form $a(p,q)$. If we do not require $a(\cdot,\cdot)$ to be symmetric, the variational problem

$$a(p,q) = \int_\Omega f q \, dx, \quad \forall q \in H_0^1(\Omega), \tag{1.3.76}$$

has for $f \in L^2(\Omega)$ a unique solution $p \in H_0^1(\Omega)$ but does not correspond to the minimisation of any functional. To fix ideas, let us suppose that $a(p,q)$ can be written as

$$a(p,q) = m(\underline{\text{grad}}\, p, \underline{\text{grad}}\, q) = \int_\Omega M(\underline{\text{grad}}\, p) \cdot \underline{\text{grad}}\, q \, dx \tag{1.3.77}$$

where $m(\cdot,\cdot)$ is a continuous bilinear form on $(L^2(\Omega))^2$, which, of course, is non symmetric, and M is the associated linear operator from $(L^2(\Omega))^2$ into $(L^2(\Omega))^2$. We can now introduce the auxiliary variable

$$\underline{u} = M(\underline{\text{grad}}\, p) \tag{1.3.78}$$

and write problem (1.3.67) in the form

$$\begin{cases} \displaystyle\int_\Omega \underline{u} \cdot \underline{\text{grad}}\, q \, dx = \int_\Omega f q \, dx, \\ \displaystyle\int_\Omega M^{-1}\underline{u} \cdot \underline{v} \, dx = \int_\Omega \underline{v} \cdot \underline{\text{grad}}\, p \, dx. \end{cases} \tag{1.3.79}$$

This can be integrated by parts to yield, as in Example 1.3.5, for $\underline{u}$ in $H(\text{div};\Omega)$ and p in $L^2(\Omega)$:

$$\begin{cases} \displaystyle\int_\Omega \text{div}\, \underline{u}\, q \, dx + \int_\Omega f q \, dx = 0, \quad \forall q \in L^2(\Omega), \\ \displaystyle\int_\Omega M^{-1}\underline{u} \cdot \underline{v} \, dx + \int_\Omega p \ \text{div}\, \underline{v} \, dx = 0, \quad \forall \underline{v} \in H(\text{div};\Omega). \end{cases} \tag{1.3.80}$$

We shall thus consider in Chaps. 3 and 4 problems such as (1.3.80) without making reference to a saddle point problem. The same remark would apply to the methods of the following section. □

1.3.4 Mixed Eigenvalue Problems

We have considered earlier in Sect. 1.2.1 some eigenvalue problems associated with our examples. We shall now rapidly consider their counterpart in mixed form. Let

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us thus come back to problem (1.3.13). It is now possible to consider three distinct eigenvalue problems.

1. The primal eigenvalue problem,

$$\begin{cases} a(u,v) + b(v,p) = \lambda (u,v)_V, & \forall v \in V, \\ b(u,q) = 0, & \forall q \in Q. \end{cases} \tag{1.3.81}$$

2. The dual eigenvalue problem,

$$\begin{cases} a(u,v) + b(v,p) = 0, & \forall v \in V, \\ b(u,q) = -\lambda (p,q)_Q, & \forall q \in Q. \end{cases} \tag{1.3.82}$$

3. The global eigenvalue problem,

$$\begin{cases} a(u,v) + b(v,p) = \lambda (u,v)_V, & \forall v \in V, \\ b(u,q) = -\lambda (p,q)_Q, & \forall q \in Q. \end{cases} \tag{1.3.83}$$

In practice, the interesting cases, from the physical point of view, will be either the primal or the dual problem.

Example 1.3.9 (Eigenvalue problem for incompressible elasticity and the Stokes problem). This case is the simplest instance of a primal eigenvalue problem of type (1.3.81). We consider the eigenvalue problem corresponding to (1.3.39) in the limiting case of λ infinitely large.

$$\begin{cases} 2\mu \displaystyle\int_\Omega \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v})\, dx + \int_\Omega p \ \operatorname{div} \underline{v}\, dx = \lambda \int_\Omega \underline{u} \cdot \underline{v}\, dx, & \forall \underline{v} \in V, \\ \displaystyle\int_\Omega q \ \operatorname{div} \underline{u} = 0, & \forall q \in L^2(\Omega). \end{cases} \tag{1.3.84}$$

This problem is the equivalent of (1.2.60) in the incompressible case. The Lagrange multiplier p ensures the incompressibility of the eigenmodes. □

Example 1.3.10 (Eigenvalue problem for the mixed Poisson problem). This is the simplest example of a dual problem of type (1.3.82). We consider the eigenvalue problem associated with the saddle-point problem of (1.3.51).

$$\begin{cases} \displaystyle\int_\Omega \underline{u} \cdot \underline{v}\, dx + \int_\Omega p \ \ \operatorname{div} \underline{v}\, dx = 0, \ \forall \underline{v} \in H(\operatorname{div}; \Omega), \\ \displaystyle\int_\Omega \operatorname{div} \underline{u}\, q\, dx = -\lambda \int_\Omega p\, q\, dx, & \forall q \in L^2(\Omega). \end{cases} \tag{1.3.85}$$

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This corresponds to the standard eigenvalue problem of (1.2.58). The Lagrange multiplier p is now the important variable. □

As we shall see later in Chap. 6, the approximation of those two kinds of problems will need specific assumptions.

1.4 Domain Decomposition Methods, Hybrid Methods

We have shown in Sect. 1.3 that duality techniques enable us to obtain alternate variational formulations for some problems. The method that we shall now describe will yield a new family of variational principles that can be more or less grouped under the name of hybrid methods. The common point between the examples that follow is that in all cases the variational principle will depend explicitly, independently of any discretisation, on a partition of the domain Ω into sub-domains. To clarify some of the facts that will appear later, we first recall a very classical result.

Example 1.4.1 (A transmission problem). We consider the very classical case in which a domain Ω is split into two sub-domains Ω_1 and Ω_2 by a smooth enough internal boundary S (Fig. 1.1). We consider the case of a Dirichlet problem with variable coefficients $a_1(x)$, $a_2(x)$, defined respectively in Ω_1 and Ω_2 and being discontinuous on S. This classically leads to the variational problem: *find* $p \in H_0^1(\Omega)$ *such that*

$$\int_{\Omega_1} a_1(x)\ \underline{\mathrm{grad}}\, p \cdot\ \underline{\mathrm{grad}}\, q\, dx + \int_{\Omega_2} a_2(x)\ \underline{\mathrm{grad}}\, p \cdot\ \underline{\mathrm{grad}}\, q\, dx = \int_{\Omega} fq\, dx, \quad \forall q \in H_0^1(\Omega). \tag{1.4.1}$$

We would like to decouple the above problem into two problems, one in each Ω_i, and add suitable continuity conditions at the interface S. For this we recall the following classical result.

Proposition 1.4.1. *Assume that Ω is a domain in $\mathbb{R}^2$ with a Lipschitz continuous boundary, and let S be a Lipschitz continuous curve that splits Ω in the two sub-domains Ω_1 and Ω_2. Let moreover $a(x)$ be a piecewise smooth function, and denote by $a_i(x)$ $(i = 1, 2)$ the restriction of $a(x)$ to Ω_i. Let $f \in L^2(\Omega)$ and let p be solution of the problem (1.4.1) Then, setting $p_1 = p|_{\Omega_1}$ and $p_2 = p|_{\Omega_2}$, it is equivalent to say that p is solution of the problem*

$$\begin{cases} -\ div\,(a_1(x)\ \underline{grad}\, p_1) = f \ in\ \Omega_1, \\ -\ div\,(a_2(x)\ \underline{grad}\, p_2) = f \ in\ \Omega_2, \\ p_1|_{\Gamma\cap\partial\Omega_1} = 0,\ p_2|_{\Gamma\cap\partial\Omega_2} = 0, \end{cases} \tag{1.4.2}$$

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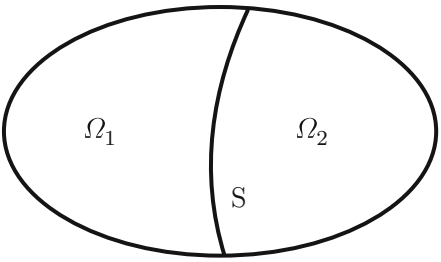


Fig. 1.1 The decomposed domain

$$p_1 = p_2 \text{ on } S,\ a_1 \frac{\partial p_1}{\partial n_1} + a_2 \frac{\partial p_2}{\partial n_2} = 0 \text{ on } S, \tag{1.4.3}$$

where n_1 and n_2 are the exterior normals to Ω_1 and Ω_2 (respectively) on S. □
□

An important special case is $a_1(x) = a_2(x) = 1$. In that case, problem (1.4.1) is obviously equivalent to

$$\begin{cases} -\Delta p = f, \\ u|_\Gamma = 0 \end{cases} \tag{1.4.4}$$

and conditions (1.4.2) and (1.4.3) can be written as

$$\begin{cases} -\Delta p_1 = f \text{ in } \Omega_1, \\ -\Delta p_2 = f \text{ in } \Omega_2, \\ p_1|_{\Gamma \cap \partial\Omega_1} = 0,\ p_2|_{\Gamma \cap \partial\Omega_2} = 0, \end{cases} \tag{1.4.5}$$

and

$$\begin{cases} p_1 = p_2 \text{ on } S, \\ \dfrac{\partial p_1}{\partial n_1} + \dfrac{\partial p_2}{\partial n_2} = 0 \text{ on } S. \end{cases} \tag{1.4.6}$$

Example 1.4.2 (A domain decomposition method for the Dirichlet problem). What we really want to do is to consider a general partition of Ω

$$\bar{\Omega} := \bigcup_{i=1}^{N} \bar{K}_i. \tag{1.4.7}$$

We now write the classical Dirichlet functional of Example 1.2.1 in the following way. First, we write the Dirichlet functional as

$$J(q) := \sum_{i=1}^{N} \Big\{ \frac{1}{2} \int_{K_i} |\ \underline{\text{grad}}\, q|^2\, dx - \int_{K_i} fq\, dx \Big\}. \tag{1.4.8}$$

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Then, introducing the functional space

$$X(\Omega) := \{q|\ q|_{K_i} \in H^1(K_i)\} \approx \prod_{i=1}^{N} H^1(K_i), \tag{1.4.9}$$

we can extend $J(q)$ on $X(\Omega)$. Moreover, $H_0^1(\Omega)$ is a closed subspace of $X(\Omega)$ and we may consider "$q \in H_0^1(\Omega)$" as a linear constraint on $q \in X(\Omega)$. This constraint states that on $e_{ij} = \partial K_i \cap \partial K_j$ we must have (in $H^{\frac{1}{2}}(e_{ij})$) $p_i = p_j$, where $p_\ell = p|_{K_\ell}$.

We shall therefore, following a now familiar procedure, impose this constraint through a Lagrange multiplier properly chosen in $H^{-\frac{1}{2}}(e_{ij})$. As we shall see in Chap. 2, it will be more convenient to introduce $\underline{v} \in H(\mathrm{div};\Omega)$ and to use the normal trace of $\underline{v}$ on ∂K_i as a multiplier. This leads us to the saddle point problem

$$\inf_{p\in X(\Omega)} \sup_{\underline{v}\in H(\mathrm{div};\Omega)} \sum_{i=1}^{N} \Big\{ \frac{1}{2} \int_{K_i} |\ \underline{\mathrm{grad}}\, q|^2\, dx - \int_{\partial K_i} \underline{v}\cdot\underline{n}_i\, q\, ds - \int_{K_i} fq\, dx \Big\}, \tag{1.4.10}$$

for which we have the following optimality conditions: for $i = 1, \ldots, N$, find $p_i \in H^1(K_i)$ such that,

$$\int_{K_i} \underline{\mathrm{grad}}\, p_i \cdot \underline{\mathrm{grad}}\, q_i\, dx = \int_{K_i} f q_i\, dx + \int_{\partial K_i} \underline{u}\cdot\underline{n}\, q_i\, ds, \quad \forall q_i \in H^1(K_i), \tag{1.4.11}$$

$$\sum_{i=1}^{N} \int_{\partial K_i} \underline{v}\cdot\underline{n}_i\ p_i\, ds = 0, \quad \forall \underline{v} \in H(\mathrm{div};\Omega). \tag{1.4.12}$$

Condition (1.4.12) expresses continuity of p at interfaces e_{ij} and condition $p|_\Gamma = 0$. Condition (1.4.11) shows that each p_i is solution in K_i of a Neumann problem

$$\begin{cases} -\Delta p_i = f \text{ in } K_i, \\ \dfrac{\partial p_i}{\partial n_i} = \underline{u}\cdot\underline{n}_i \text{ on } \partial K_i. \end{cases} \tag{1.4.13}$$

Solving this problem obviously requires a compatibility condition (take $q_i = 1$ in (1.4.11))

$$\int_{\partial K_i} \underline{u}\cdot\underline{n}_i\, ds + \int_{K_i} f\, dx = 0 \tag{1.4.14}$$

on every sub-domain K_i. This condition can also be written as

$$\int_{K_i} (\mathrm{div}\ \underline{u} + f)\, dx = 0. \tag{1.4.15}$$

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From (1.4.13) we have that the multiplier $\underline{u} \cdot \underline{n}$ can be seen as the normal derivative of p. Indeed, when equilibrium is attained, we have on interfaces $\frac{\partial p_i}{\partial n_i} = \underline{u} \cdot \underline{n}_i = -\underline{u} \cdot \underline{n}_j = \frac{-\partial p_j}{\partial n_j}$ and $p_i = p_j$. A suitable lifting of $\underline{u}$ in each K_i in order to have div $\underline{u} + f = 0$ can always be done because of (1.4.14) and (1.4.15). □

Example 1.4.3 (Dual problem of the domain decomposition method). We now consider the dual problem of the above saddle point formulation. It will be, as it can be expected, very similar to the dual problem introduced in Sect. 1.3 for the Poisson problem. Let us first remark that taking the *infimum* on the constant part of $q \in X(\Omega)$ on each K_i leads to the *constraint* (1.4.15) on $\underline{u}$. It is therefore possible to suppose div $\underline{u} + f = 0$ as this can be attained by modifications to $\underline{u}$ that are internal to K_i (that is not modifying $\underline{u} \cdot \underline{n}_i$) and are transparent to formulation (1.4.10). Writing

$$\int_{\partial K_i} \underline{v} \cdot \underline{n}_i \; q \; ds = \int_{K_i} \text{div } \underline{u} \; q \; dx + \int_{K_i} \underline{v} \cdot \; \underline{\text{grad}}\, q \; dx, \tag{1.4.16}$$

one may write from (1.4.10)

$$\sup_{\text{div}\,(\underline{v})+f=0} \; \inf_{q_i \in H^1(K_i)/\mathbb{R}} \sum_{i=1}^{N} \Big\{ \frac{1}{2} \int_{K_i} | \; \underline{\text{grad}}\, q_i |^2 \; dx - \int_{K_i} \underline{v} \cdot \; \underline{\text{grad}}\, q_i \; dx \Big\}. \tag{1.4.17}$$

From (1.4.17) we evidently get, setting $\underline{u}_i = \underline{u}|_{K_i}$,

$$\underline{\text{grad}}\; p_i = P(\underline{u}_i), \tag{1.4.18}$$

where P is the projection operator in $(L^2(K_i))^2$ on $\underline{\text{grad}}(H^1(K_i))$. We shall indeed prove in Chap. 2 that one has

$$(L^2(\Omega))^2 = \{\underline{\text{grad}}\, H^1(\Omega)\} \oplus \underline{\text{curl}}\, H_0^1(\Omega)\}. \tag{1.4.19}$$

From this we can eliminate q_i and write the dual problem,

$$\sup_{\substack{\underline{v} \in H(\text{div};\Omega) \\ \text{div}(\underline{v})+f=0}} -\frac{1}{2} \sum_{i=1}^{N} \int_{K_i} |P(\underline{v}_i)|^2 \; dx. \tag{1.4.20}$$

We are therefore back to a variant of (1.3.45). Indeed, (1.3.45) shows that the projection operator P in (1.4.20) is unnecessary. □

Remark 1.4.1. One could obtain a variant of the above dual problem, without constraint (1.4.15) by using a "least-squares" solution of (1.4.13) whenever (1.4.14) does not hold. This could be done, for instance by solving on K_i, in a weak formulation that we shall not describe,

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$$\begin{cases} \Delta^2 p_i = \Delta f \text{ in } K_i, \\ \dfrac{\partial}{\partial n_i}\Delta p_i = \dfrac{\partial f}{\partial n_i} \text{ on } \partial K_i, \\ \dfrac{\partial p_i}{\partial n_i} = \underline{v}\cdot\underline{n}_i \text{ on } \partial K_i, \end{cases} \tag{1.4.21}$$

for which a solution always exists, defined up to an additive constant. Such a procedure could be useful for algorithmic purposes for (1.4.21) is a local simple problem even if it is a fourth-order problem. □

Example 1.4.4 (Dual hybrid methods). We now consider the dual problem (1.3.45), that is the complementary energy principle, that we now pose in $H(\mathrm{div};\Omega)$,

$$\inf_{\substack{\underline{v}\in H(\mathrm{div};\Omega)\\ \mathrm{div}\,(\underline{v})+f=0}} \frac{1}{2}\int_\Omega |\underline{v}|^2\,dx. \tag{1.4.22}$$

We can apply the domain decomposition principle to such a problem by introducing

$$Y(\Omega) := \{\underline{v}|\ \underline{v}|_{K_i} \in H(\mathrm{div};K_i)\} \approx \prod_{i=1}^N H(\mathrm{div};K_i). \tag{1.4.23}$$

As we shall see in Chap. 2, $H(\mathrm{div};\Omega)$ is now a closed subspace of $Y(\Omega)$ characterised by

$$\sum_{i=1}^N \int_{\partial K_i} (\underline{v}\cdot\underline{n}_i)\,q\,ds = 0, \quad \forall q \in H_0^1(\Omega). \tag{1.4.24}$$

We can then transform (1.4.22) into the saddle point problem

$$\inf_{v\in Y(\Omega)} \sup_{q\in H_0^1(\Omega)} \sum_{i=1}^N \left\{\frac{1}{2}\int_{K_i} |\underline{v}_i|^2\,dx + \int_{\partial K_i} \underline{v}_i\cdot\underline{n}_i\,q\,ds\right\} \tag{1.4.25}$$

under the local constraint

$$\mathrm{div}\ \underline{v}_i + f = 0 \text{ on } K_i. \tag{1.4.26}$$

An advantage of this formulation is that it is easy to find $\underline{v}_i$ satisfying (1.4.26). We shall meet discretisation methods, based on such a principle, under the name of *dual hybrid methods* for the treatment of almost any example considered in this book: Dirichlet problems, elasticity problems, fourth-order problems, etc. □

Example 1.4.5 (The Hellan-Hermann-Johnson method in elasticity). This is an example in which a domain decomposition is introduced, not by dualising a

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continuity condition but by defining a variational formulation able to bypass this continuity by approximating weak derivatives. This formulation will not be developed but is amenable to the techniques of the book. We shall first present formal results and postpone a precise presentation of the functional framework. Our starting point will be the saddle point problem (1.3.59) and its optimality conditions (1.3.60) that we write, in variational form (with functional spaces to be defined), as

$$\frac{1}{\mu}\int_\Omega \underline{\underline{\sigma}}^D : \underline{\underline{\tau}}^D\, dx + \frac{1}{2(\lambda+\mu)}\int_\Omega tr\,\underline{\underline{\sigma}}\; tr\,\underline{\underline{\tau}}\, dx + \int_\Omega \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{\tau}}\, dx = 0, \quad \forall \underline{\underline{\tau}} \in \underline{\underline{H}}(\underline{\mathrm{div}};\Omega)_s, \tag{1.4.27}$$

$$\int_\Omega \underline{\underline{\varepsilon}}(\underline{v}) : \underline{\underline{\sigma}}\, dx + \int_\Omega \underline{f}\cdot\underline{v}\, dx = 0 \qquad \forall \underline{v} \in (H_0^1(\Omega))^2. \tag{1.4.28}$$

These conditions make sense for a space of $\underline{\underline{\sigma}}$ chosen so that $\underline{\mathrm{div}}\,\underline{\underline{\sigma}}$ is well defined, which implies, as we have seen, continuity of $\underline{\sigma}_n$ at interfaces. On the other hand $\underline{v}$ can be taken as completely discontinuous on these same interfaces. What we now try to do is to split continuity conditions between $\underline{\sigma}\cdot\underline{n}$ and $\underline{v}$. Let us consider indeed the well-known integration by parts formula,

$$\int_\Omega \underline{\mathrm{div}}\,\underline{\underline{\tau}}\cdot\underline{v}\, dx + \int_\Omega \underline{\underline{\tau}} : \underline{\underline{\varepsilon}}(\underline{v})\, dx = \int_{\partial\Omega} \tau_{nn}\,\underline{v}\cdot\underline{n}\, ds + \int_{\partial\Omega} \tau_{nt}\,\underline{v}\cdot\underline{t}\, ds. \tag{1.4.29}$$

Whenever $\underline{v}$ is a smooth (let us say $H^1(\Omega)$) vector, and τ_{nt} is continuous, we thus have

$$\int_\Omega \underline{\underline{\varepsilon}}(\underline{v}) : \underline{\underline{\sigma}}\, dx = -\sum_{i=1}^N \left\{ \int_{K_i} \underline{\mathrm{div}}\,\underline{\underline{\sigma}}\cdot\underline{v}\, dx + \int_{\partial K_i} \sigma_{nn}\,\underline{v}\cdot\underline{n}\, ds \right\}, \tag{1.4.30}$$

so that we can rewrite (1.4.27) and (1.4.28) in the following form,

$$\frac{1}{\mu}\int_\Omega \underline{\underline{\sigma}}^D : \underline{\underline{\tau}}^D\, dx + \frac{1}{2(\lambda+\mu)}\int_\Omega tr\,\underline{\underline{\sigma}}\; tr\,\underline{\underline{\tau}}\, dx + \sum_{i=1}^N \left\{ \int_{K_i} \underline{\mathrm{div}}\,\underline{\underline{\tau}}\cdot\underline{u}\, dx - \int_{\partial K_i} \tau_{nn}\,\underline{u}\cdot\underline{n}\, ds \right\} = 0 \quad \forall \underline{\underline{\tau}}, \tag{1.4.31}$$

$$\sum_{i=1}^N \left\{ \int_{K_i} \underline{\mathrm{div}}\,\underline{\underline{\sigma}}\cdot\underline{v}\, dx - \int_{\partial K_i} \sigma_{nn}\,\underline{v}\cdot\underline{n}\, ds \right\} + \int_\Omega \underline{f}\cdot\underline{v}\, dx = 0 \quad \forall \underline{v}. \tag{1.4.32}$$

Formally, this is well defined for $\underline{\underline{\sigma}}$ chosen with σ_{nt} continuous at interfaces while $\underline{u}\cdot\underline{n}$ is continuous. Then the term

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$$\sum_{i=1}^{N}\Big\{\int_{\partial K_i}\sigma_{nn}\,\underline{v}\cdot\underline{n}\,ds\Big\} \tag{1.4.33}$$

depends on the *jump* of σ_{nn} on ∂K_i and (1.4.32) can be read as $\underline{\underline{\mathrm{div}}}\,\underline{\underline{\sigma}}+\underline{f}=0$ in the sense of distributions.

Up to now we considered a purely formal problem. Giving a good framework to (1.4.31), and (1.4.32) is a task that requires some care. The presence of traces, appearing explicitly in the variational formulation, leads to deal with spaces $H^{\frac{1}{2}}(\partial K_i)$ and $H^{-\frac{1}{2}}(\partial K_i)$ and to subtle considerations about the behaviour of functions in these rather pathological spaces. Let us define

$$\Sigma:=\prod_{K_i}(H^1(K_i))^4_s=\{\underline{\underline{\sigma}}|\ \sigma_{ij}|_{K_i}\in H^1(K_i),\ \sigma_{ij}=\sigma_{ji}\}. \tag{1.4.34}$$

This is a space of smooth tensors and we can consider σ_{nt} on each interface $e_{ij}=\partial K_i\cap\partial K_j$, (cf. Chap. 2). We have $\sigma_{nt}\in H^{\frac{1}{2}}(e_{ij})$ but we do not have $\sigma_{nt}\in H^{\frac{1}{2}}(\partial K)$ for this would require some continuity at vertices which cannot in general take place due to the change of direction of $\underline{n}$ and $\underline{t}$. We can nevertheless consider in Σ tensor functions $\underline{\underline{\sigma}}$ such that σ_{nt} is continuous on e_{ij}. To make (1.4.33) meaningful, we now have to choose $\underline{v}$ with $\underline{v}\cdot\underline{n}$ continuous on e_{ij}. We have already seen that for $\underline{v}$ in $H(\mathrm{div};K_i)$ we can define $\underline{v}\cdot\underline{n}$ in $H^{-\frac{1}{2}}(\partial K_i)$. Unfortunately it is not possible to restrict $\underline{v}\cdot\underline{n}|_{e_{ij}}$ and get a result in $H^{-\frac{1}{2}}(e_{ij})$: something is lost at corners. In reality we only need an "infinitesimal" amount of extra smoothness and this will lead us to look for $\underline{v}$ in $(L^p(\Omega))^2\cap H(\mathrm{div};\Omega)$ for $p>2$. This will cause some problems in applying the theory of Chap. 4 and existence of a solution will have to be deduced through special considerations. □

1.5 Modified Variational Formulations

We shall present in this section modified variational principles associated to saddle-point problems or more general mixed methods. We shall distinguish between augmented formulations and perturbed formulations. In the following, *augmented formulations* will correspond to a modification of the variational formulation of the **continuous** problem. This will be done so that the solution is not changed (under perhaps some regularity conditions). The discretised versions will however in general be different. On the other hand, perturbed formulations will be meaningful only on a discrete problem. The rationale behind the introduction of these modified formulations is that they will have, for some discretisations, a better behaviour than the original one, in particular with respect to stability issues.

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1.5.1 Augmented Formulations

As an example, let us consider the Galerkin least-squares methods introduced by Hughes and Franca [256]. To fix ideas, let us consider the simple cases of the saddle point formulation of the Poisson problem of Sect. 1.3,

$$\inf_{\underline{v}\in H(\mathrm{div};\Omega)} \sup_{q\in L^2(\Omega)} \frac{1}{2}\int_\Omega |\underline{v}|^2\,dx + \int_\Omega fq\,dx + \int_\Omega q\ \ \mathrm{div}\ \underline{v}\,dx, \tag{1.5.1}$$

for which the Euler equations are

$$\begin{cases} \displaystyle\int_\Omega \underline{u}\cdot\underline{v}\,dx + \int_\Omega p\ \ \mathrm{div}\ \underline{v}\,dx = 0 & \forall \underline{v}\in H(\mathrm{div};\Omega), \\ \displaystyle\int_\Omega (\mathrm{div}\ \underline{u} + f)q\,dx = 0 & \forall q\in L^2(\Omega). \end{cases} \tag{1.5.2}$$

To better understand stability issues, it will be convenient to write (1.5.2) using an antisymmetric bilinear form, obtained by subtracting the two equations

$$\begin{aligned} E((p,\underline{u}),(q,\underline{v})) :&= \int_\Omega \underline{u}\cdot\underline{v}\,dx + \int_\Omega p\ \ \mathrm{div}\ \underline{v}\,dx - \int_\Omega q\ \ \mathrm{div}\ \underline{u}\,dx \\ &= \int_\Omega fq\,dx, \qquad \forall \underline{v}\in H(\mathrm{div};\Omega), \forall q\in L^2(\Omega). \end{aligned} \tag{1.5.3}$$

One sees that

$$E((p,\underline{u}),(p,\underline{u})) = \int_\Omega |\underline{u}|^2\,dx, \tag{1.5.4}$$

so that our bilinear form is non-negative, but not coercive.

The approximation of (1.5.1) thus requires the special constructions that will be described in Chaps. 3–5.

Remark 1.5.1. **A basic philosophical issue:** It is important, at this point, to underline a basic philosophical issue: Assume that you are given a bilinear form (say, E) on $\mathcal{H}\times\mathcal{H}$, where $\mathcal{H}$ is a Hilbert space. There are several ways that could be used to prove that *E induces an isomorphism from $\mathcal{H}$ to its dual space $\mathcal{H}'$*. We mean by that that there are several properties that will imply such a result. For instance, if for simplicity $\mathcal{H}\equiv\mathbb{R}^n$, you can show that the associated matrix M_E has a determinant that is different from zero. Otherwise, you can show that the associated homogeneous system has only the (trivial) zero solution. Alternatively, you can show that the associated non-homogeneous system has at least one solution for every right-hand side, or you can show that there exists a constant c such that for every pair (X,F) that satisfies $M_E X = F$ you have $\|X\| \le c\,\|F\|$. All

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these properties are indeed *equivalent* to each other (in finite dimension). As a last possibility, you can show that E is *coercive*:

$$E(X, X) \geq \alpha \|X\|^2 \qquad \forall X \in \mathcal{H}. \tag{1.5.5}$$

This last condition (some sort of Cinderella among the other equivalent step-sisters) is not *necessary and sufficient* (as all the others are), but only *sufficient*. Indeed, there are zillions of non-singular matrices that are not coercive. Assume now that you want to play an additional game. You would like to consider *proper subspaces* $\tilde{\mathcal{H}}$ of $\mathbb{R}^n$ (say, to fix ideas, $\mathbb{R}^m$ with $m < n$) and the *restriction* $\tilde{E}$ of the bilinear form E to $\tilde{\mathcal{H}} \times \tilde{\mathcal{H}}$. We ask whether the bilinear form $\tilde{E}$ induces an isomorphism from $\tilde{\mathcal{H}}$ to its dual $\tilde{\mathcal{H}}'$. In fact, not a single one of the above necessary and sufficient conditions will be *automatically inherited* by $\tilde{E}$. Actually, as they are all equivalent, if one does not the others cannot do either. Surely there are zillions of non-singular matrices M whose first entry $M_{1,1}$ is equal to 0. Then you take $\tilde{\mathcal{H}}$ equal to $\mathbb{R}^1$ (using the first component of every $X \in \mathcal{H} \equiv \mathbb{R}^n$) and you are done: $\tilde{E}$ is 0.

But Cinderella *survives*: if E is coercive on $\mathcal{H}$, then $\tilde{E}$ is coercive on $\tilde{\mathcal{H}}$, *for every subspace* $\tilde{\mathcal{H}} \subseteq \mathcal{H}$ and the same value of α that makes (1.5.5) true will make

$$\tilde{E}(\tilde{X}, \tilde{X}) \geq \alpha \|\tilde{X}\|^2 \qquad \forall \tilde{X} \in \tilde{\mathcal{H}}, \tag{1.5.6}$$

hold true as well. Hence, Cinderella becomes princess and superstar, and everybody, for every problem, would like to have a coercive bilinear form. If it is not, one would struggle to change the problem into an equivalent one, whose associated bilinear form is *coercive*. This, in short, is the essence of many stabilisation techniques. □

We have already seen in Sect. 1.3 that the solution of the "weaker" problem (1.5.2) is in fact the solution of the standard problem $-\Delta p = f$, written as the system

$$\begin{cases} \underline{u} - \underline{\text{grad}}\, p = 0, & p \in H_0^1(\Omega), \\ \text{div}\, \underline{u} + f = 0, & \underline{u} \in H(\text{div}; \Omega). \end{cases} \tag{1.5.7}$$

Starting from this system, we can also consider the other formulation

$$\inf_{\underline{v} \in (L^2(\Omega))^2} \sup_{q \in H_0^1(\Omega)} \frac{1}{2} \int_\Omega |\underline{v}|^2 \, dx + \int_\Omega fq \, dx - \int_\Omega \underline{\text{grad}}\, q \cdot \underline{v} \, dx, \tag{1.5.8}$$

for which the Euler equations are now

$$\begin{cases} \displaystyle\int_\Omega \underline{u} \cdot \underline{v} \, dx - \int_\Omega \underline{\text{grad}}\, p \cdot \underline{v} \, dx = 0, & \forall \underline{v} \in (L^2(\Omega))^2, \\ \displaystyle -\int_\Omega \underline{u} \cdot \underline{\text{grad}}\, q \, dx + \int_\Omega fq \, dx = 0, & \forall q \in H_0^1(\Omega). \end{cases} \tag{1.5.9}$$

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Note that, comparing to (1.5.2), we had to change the regularity requirements on q in order to make the functional meaningful. One checks that the solution of the continuous Poisson problem, which belongs to $H_0^1(\Omega)$, is a solution of our new saddle point problem. However, it is now clear that the discrete problems will now have to employ finite element approximations of $H_0^1(\Omega)$.

It is clear that we can always add (or subtract) the square of one of the equations (1.5.7) to the Lagrangian of (1.5.1) or (1.5.8) without changing the min-max point. For instance, we can add to (1.5.1) the square of the second equation of (1.5.7) to obtain

$$\inf_{\underline{v}\in H(\mathrm{div};\Omega)} \sup_{q\in L^2(\Omega)} \left\{ \frac{1}{2}\int_\Omega |\underline{v}|^2\,dx + \int_\Omega fq\,dx + \int_\Omega q\ \mathrm{div}\,\underline{v}\,dx + \frac{\kappa_1}{2}\int_\Omega (\mathrm{div}\,\underline{v}+f)^2\,dx \right\}, \quad (1.5.10)$$

where $\kappa_1 \geq 0$ is arbitrary. We write again the Euler equations using an antisymmetric bilinear form,

$$\begin{aligned} E((p,\underline{u}),(q,\underline{v})) &= \int_\Omega \underline{u}\cdot\underline{v}\,dx + \kappa_1\int_\Omega \mathrm{div}\ \underline{u}\ \mathrm{div}\ \underline{v}\,dx \\ &\quad + \int_\Omega p\ \ \mathrm{div}\ \underline{v}\,dx - \int_\Omega q\ \mathrm{div}\ \underline{u}\,dx \\ &= \int_\Omega fq\,dx - \kappa_1\int_\Omega f\ \ \mathrm{div}\ \underline{v}\,dx, \\ &\qquad \forall \underline{v}\in H(\mathrm{div};\Omega), \forall q\in L^2(\Omega). \end{aligned} \quad (1.5.11)$$

It is clear that the solution of problem (1.5.10) is exactly the same as that of problem (1.5.1). Approximate solutions might however be different and some choices of elements will be stable for (1.5.10) but not for (1.5.1). Indeed, we now have

$$\begin{aligned} E((q,\underline{v}),(q,\underline{v})) &= \int_\Omega |\underline{v}|^2\,dx + \kappa_1\int_\Omega |\ \mathrm{div}\,\underline{v}|^2\,dx \\ &\geq \min(1,\kappa_1)\|\underline{v}\|^2_{H(\mathrm{div};\Omega)}, \end{aligned} \quad (1.5.12)$$

which is not yet the coercivity property (1.5.5), but is much better than (1.5.4), as now at least the full norm of $\underline{v}$ is under control. Indeed, as we shall see, this enables the construction of otherwise impossible approximations such as in [124].

Remark 1.5.2. Terms like $\min(1,\kappa_1)$ appearing in (1.5.12) are also *weird*. Their presence has still to do with the weird choice of the norm (1.3.47) in $H(\mathrm{div};\Omega)$. Had we chosen the more reasonable (but, alas!, almost never used in the literature) definition (1.3.49), we would, instead, have reached a $\min(1,\kappa_1/\ell)$ which looks *much more healthy*, as κ_1 is clearly a length. □

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Another reasonable possibility is available: one might take (1.5.8) and subtract from it the square of the first equation of (1.5.7), to get

$$\inf_{\underline{v}\in(L^2(\Omega))^2}\sup_{q\in H_0^1(\Omega)}\frac{1}{2}\int_\Omega|\underline{v}|^2\,dx+\int_\Omega fq\,dx-\int_\Omega \underline{\mathrm{grad}}\,q\,\underline{v}\,dx$$
$$-\frac{\kappa_2}{2}\int_\Omega|\underline{v}-\underline{\mathrm{grad}}\,q|^2dx \quad (1.5.13)$$

and the Euler equations of this Lagrangian yield

$$E((p,\underline{u}),(q,\underline{v}))=(1-\kappa_2)\int_\Omega\underline{u}\cdot\underline{v}\,dx-(1-\kappa_2)\int_\Omega\underline{\mathrm{grad}}\,p\cdot\underline{v}\,dx$$
$$+(1-\kappa_2)\int_\Omega\underline{u}\cdot\,\underline{\mathrm{grad}}\,q\,dx+\kappa_2\int_\Omega\underline{\mathrm{grad}}\,p\cdot\,\underline{\mathrm{grad}}\,q\,dx$$
$$=\int_\Omega fq\,dx\qquad\forall\underline{v}\in(L^2(\Omega))^2,\forall q\in H_0^1(\Omega). \quad (1.5.14)$$

This equation is a convex combination of equation (1.5.3) and of a standard formulation of the Poisson problem. Although this may seem silly at first sight, such methods have been used in [208] to stabilise formulations in which the discrete space V_h for the approximation of $\underline{u}$ was too small with respect to the space spanned by $\underline{\mathrm{grad}}\,q_h$, resulting in a failure of the *inf-sup* condition. The reason to employ such a strange approximation was that the first equation was much more complex than $\underline{u}-\underline{\mathrm{grad}}\,p=0$ but rather of the form $\underline{u}-\underline{\mathrm{grad}}\,p=F(\underline{u},p)$ where the function $F(\underline{u},p)$ contained terms prescribing low order elements in order to be manageable. Unsurprisingly, we now have

$$E((q,\underline{v}),(q,\underline{v}))=(1-\kappa_2)\int_\Omega|\underline{v}|^2\,dx+\kappa_2\int_\Omega|\,\underline{\mathrm{grad}}\,q|^2\,dx, \quad (1.5.15)$$

and we have coercivity on $(L^2(\Omega))^2\times H_0^1(\Omega)$ for $0<\kappa_2<1$.

Now, if one is really eager for stability, one could consider a "super-stabilised" formulation

$$\inf_{\underline{v}\in H(\mathrm{div};\Omega)}\sup_{q\in H_0^1(\Omega)}\frac{1}{2}\int_\Omega|\underline{v}|^2\,dx-\int_\Omega\underline{v}\cdot\,\underline{\mathrm{grad}}\,q\,dx+\int_\Omega fq\,dx$$
$$+\frac{\kappa_2}{2}\int_\Omega|\underline{v}-\underline{\mathrm{grad}}\,q|^2dx+\frac{\kappa_1}{2}\int_\Omega(\mathrm{div}\,\underline{v}+f)^2\,dx$$
$$+\frac{\kappa_3}{2}\|-\Delta q-f\|_*^2, \quad (1.5.16)$$

where $\|\cdot\|_*$ is the norm in $H^{-1}(\Omega)$, which we can define from the corresponding scalar product:

$$\langle p',q'\rangle_*:=\langle(-\Delta)^{-1}p',q'\rangle_{H_0^1(\Omega)\times H^{-1}(\Omega)}. \quad (1.5.17)$$

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Taking the Euler equation and using

$$\langle -\Delta p - f, -\Delta q\rangle_* = \langle -\Delta p - f, q\rangle_{H^{-1}(\Omega)\times H_0^1(\Omega)} = \int_\Omega \underline{\mathrm{grad}}\, p \cdot \underline{\mathrm{grad}}\, q\, dx - \int_\Omega fq\, dx,$$

one obtains

$$\begin{aligned}
&(1+\kappa_2)\int_\Omega \underline{u}\cdot\underline{v}\, dx + \kappa_1 \int_\Omega \mathrm{div}\, \underline{u}\ \mathrm{div}\, \underline{v}\, dx - (1+\kappa_2)\int_\Omega \underline{\mathrm{grad}}\, p \cdot \underline{v}\, dx \\
&= -\kappa_1 \int_\Omega f\ \mathrm{div}\, \underline{v}\, dx, \qquad \forall \underline{v} \in H(\mathrm{div};\Omega) \\
&-(1+\kappa_2)\int_\Omega \underline{u}\cdot\ \underline{\mathrm{grad}}\, q\, dx + (\kappa_2+\kappa_3)\int_\Omega \underline{\mathrm{grad}}\, p \cdot\ \underline{\mathrm{grad}}\, q\, dx \\
&= (\kappa_3 - 1)\int_\Omega fq\, dx, \qquad \forall q \in H_0^1(\Omega).
\end{aligned} \tag{1.5.18}$$

Remark 1.5.3. If we take $\kappa_3 = 1,\ \kappa_2 = -1/2,\ \kappa_1 = 1/2$, the above system (1.5.18) reduces to

$$\begin{aligned}
&\int_\Omega (\underline{u} - \underline{\mathrm{grad}}\, p)\cdot\underline{v}\, dx + \int_\Omega (\mathrm{div}\, \underline{u} + f)\ \mathrm{div}\, \underline{v}\, dx = 0, \quad \forall \underline{v} \in H(\mathrm{div};\Omega) \\
&-\int_\Omega (\underline{u} - \underline{\mathrm{grad}}\, p)\cdot\ \underline{\mathrm{grad}}\, q\, dx = 0, \qquad \forall q \in H_0^1(\Omega)
\end{aligned} \tag{1.5.19}$$

which are the Euler equations of the problem

$$\inf_{\underline{v}\in H(\mathrm{div};\Omega)} \inf_{q\in H_0^1(\Omega)} \frac{\kappa_2}{2}\int_\Omega |\underline{v} - \underline{\mathrm{grad}}\, q|^2 dx + \frac{\kappa_1}{2}\int_\Omega (\mathrm{div}\, \underline{v} + f)^2\, dx, \tag{1.5.20}$$

which is the least squares method introduced by Bramble et al. [108] and has given rise to a vast literature. □

Let us now consider the question of the stability of (1.5.18). This can be done in two ways. In the first one we subtract the two equations to get an antisymmetric bilinear form, $E_a((p,\underline{u}),(q,\underline{v}))$. Then, we clearly have

$$\begin{aligned}
E_a((q,\underline{v}),(q,\underline{v})) = (1+\kappa_2)\int_\Omega |\underline{v}|^2\, dx + \kappa_1 \int_\Omega |\ \mathrm{div}\, \underline{v}|^2\, dx \\
-(\kappa_2+\kappa_3)\int_\Omega |\ \underline{\mathrm{grad}}\, q|^2\, dx.
\end{aligned} \tag{1.5.21}$$

We thus get coercivity if $\kappa_1 > 0,\ (1+\kappa_2) > 0,\ (\kappa_2+\kappa_3) < 0$. It is easy to see that these conditions imply $\kappa_3 < 1$ and that, this being given, one can take $\kappa_2 = -\frac{\kappa_3+1}{2}$.

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On the other hand, one could want to consider a symmetrical bilinear form by adding the two equations instead of subtracting them. One then has

$$E_s((q,\underline{v}),(q,\underline{v})) = (1+\kappa_2)\int_\Omega |\underline{v}|^2\,dx + \kappa_1\int_\Omega |\ \mathrm{div}\,\underline{v}|^2\,dx$$
$$-2(1+\kappa_2)\int_\Omega \underline{\mathrm{grad}}\,q\cdot\underline{v}\,dx + (\kappa_2+\kappa_3)\int_\Omega |\ \underline{\mathrm{grad}}\,q|^2\,dx. \quad (1.5.22)$$

It is not difficult to check that (1.5.22) yields coercivity if

$$\kappa_1 > 0,\ \ \kappa_2 > -1,\ \ \kappa_2+\kappa_3 > 0,$$

and finally

$$(\kappa_2+\kappa_3)\min\{\kappa_1, 1+\kappa_2\} > (1+\kappa_2)^2.$$

Now, if $\min\{\kappa_1, 1+\kappa_2\} = 1+\kappa_2$, the last condition implies $\kappa_2+\kappa_3 < 1+\kappa_2$, that is $\kappa_3 < 1$.

Remark 1.5.4. As in Remark 1.5.2, instead of the expression $\min\{\kappa_1, 1+\kappa_2\}$, we should actually have $\min\{\kappa_1/\ell, 1+\kappa_2\}$ where ℓ is a characteristic length of the problem. □

Remark 1.5.5. We thus see that the least-squares formulation of Remark 1.5.3, which is obtained with $\kappa_3 = 1$, is a difficult case which cannot be studied by simple arguments and is therefore not a simple way to obtain a stable method. □

Remark 1.5.6. We presented here an example of a quite general idea which was introduced in [256] and [213]. Other examples of these ideas will be developed in the following sections of this chapter or in the following chapters. □

Remark 1.5.7. In the example presented above, Euler equations (1.5.2) were a system of first order equations. This is not the case of the Stokes problem

$$\inf_{\underline{v}\in(H_0^1(\Omega))^2}\ \sup_{q\in L^2(\Omega)}\ \mu\int_\Omega |\underline{\underline{\varepsilon}}(\underline{v})|^2 dx - \int_\Omega \underline{f}\cdot\underline{v}\,dx - \int_\Omega q\ \mathrm{div}\,\underline{v}\,dx, \quad (1.5.23)$$

for which one of the equations in strong form contains second derivatives. Applying the same procedure would lead to a fourth order problem in the variable $\underline{u}$ which would lead to undesirable complications. Indeed the analogue of (1.5.13) would here be obtained from (1.5.23) as follows:

$$\inf_{\underline{v}\in(H_0^1(\Omega))^2}\ \sup_{q\in L^2(\Omega)}\ \mu\int_\Omega |\underline{\underline{\varepsilon}}(\underline{v})|^2 dx - \int_\Omega \underline{f}\cdot\underline{v}\,dx$$
$$-\int_\Omega q\ \mathrm{div}\,\underline{v}\,dx - \beta\int_\Omega |A\underline{u} + \underline{\mathrm{grad}}\,p - \underline{f}|^2 dx, \quad (1.5.24)$$

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which would force us to use a very regular approximation for the variable $\underline{u}$ or to rather consider the formulation

$$\inf_{\underline{v}\in(H_0^1(\Omega))^2}\ \sup_{q\in L^2(\Omega)}\ \mu\int_\Omega|\underline{\underline{\dot{\varepsilon}}}(\underline{v})|^2dx-\int_\Omega\underline{f}\cdot\underline{v}\,dx$$
$$-\int_\Omega q\ \operatorname{div}\underline{v}\,dx-\beta\|A\underline{u}+\ \underline{\operatorname{grad}}\,p-\underline{f}\|_*^2,\quad(1.5.25)$$

where $\|\cdot\|_*$ denotes the norm in $H^{-1}(\Omega)$. If we take the derivative of (1.5.25) with respect to $\underline{v}$, we could write as in (1.5.17),

$$\langle A\underline{u}+\ \underline{\operatorname{grad}}\,p-\underline{f},A\underline{v}\rangle_*=\langle A\underline{u}+\ \underline{\operatorname{grad}}\,p-\underline{f},\underline{v}\rangle_{V'\times V},\quad(1.5.26)$$

but the derivative with respect to p will yield a term of the form

$$\langle A\underline{u}+\ \underline{\operatorname{grad}}\,p-\underline{f},A\underline{v}\rangle_*,$$

which is not readily computable and which will require some special handling when considering discrete approximations. The typical solution, when dealing with a decomposition of Ω into elements K and piecewise smooth functions $\underline{u}$, $\underline{v}$, p and q, is to consider terms of the form

$$\sum_K\beta_K(\operatorname{diam}(K))^2\int_K|A\underline{u}+\ \underline{\operatorname{grad}}\,p-\underline{f}|^2dx.\quad(1.5.27)$$

We shall consider solutions to this question in Chap. 8 where Stokes' problem will be studied in detail. □

Remark 1.5.8. If one looks carefully at the Euler equations of (1.5.14), one sees that we could be in some trouble for $\beta=1$ as the first equation disappears and we obviously loose control over p. This justified, in [179], the introduction of another variant of the general idea developed above. The formulation cannot in this case be written as a modified Lagrangian but is rather derived from the antisymmetric bilinear form (1.5.3). Indeed, let us write,

$$E((u,\underline{p}),(v,\underline{q}))=\int_\Omega\underline{p}\cdot\underline{q}\,dx+\int_\Omega\underline{p}\cdot\ \underline{\operatorname{grad}}\,v\,dx-\int_\Omega\underline{q}\cdot\ \underline{\operatorname{grad}}\,u\,dx$$
$$+\beta\int_\Omega(\underline{p}-\ \underline{\operatorname{grad}}\,u)\cdot(\underline{q}-\ \underline{\operatorname{grad}}\,v)dx$$
$$=-\int_\Omega fv\,dx,\qquad\forall(\underline{q},v)\in(L^2(\Omega))^2\times H_0^1(\Omega).\quad(1.5.28)$$

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This formulation cannot be obtained from a Lagrangian. It can easily be seen that it remains valid for $\beta > 0$ arbitrary. Indeed (1.5.28) can also be written as

$$(1+\beta)\int_\Omega (\underline{p} - \underline{\mathrm{grad}}\, u)\cdot \underline{q}\, dx = 0, \qquad \forall \underline{q} \in (L^2(\Omega))^2,$$
$$\beta \int_\Omega \underline{\mathrm{grad}}\, u \cdot \underline{\mathrm{grad}}\, v\, dx + (1-\beta)\int_\Omega \underline{p}\cdot \underline{\mathrm{grad}}\, v\, dx - \int_\Omega f v\, dx = 0, \quad \forall v \in H_0^1(\Omega), \tag{1.5.29}$$

and this is equivalent to (1.5.7) for any $\beta \geq 0$ with the control of $\underline{p}$ remaining untouched. Moreover, we can easily see that, as long as $\beta > 0$, there exists a $\delta > 0$ such that

$$E((v,\underline{p}),(v,\underline{q})) = \int_\Omega |\underline{q}|^2\, dx + \beta \int_\Omega |\underline{q} - \underline{\mathrm{grad}}\, v|^2\, dx > \delta \int_\Omega |\underline{q}|^2\, dx + \int_\Omega |\underline{\mathrm{grad}}\, v|^2\, dx. \tag{1.5.30}$$

We shall see later on how this technique can indeed be included in the same general framework as the previous one. □

1.5.2 *Perturbed Formulations*

We can also consider another type of modified variational formulation which is introduced in the *discretised* problems and in which, hopefully, the additional term vanishes when the mesh is refined and the numerical solution converges to the solution of the original problem. As an example, we consider the Stokes problem presented above in (1.5.23). We now suppose that we have a subspace V_h of $(H_0^1(\Omega))^2$ and Q_h of $L^2(\Omega)$ and we try to solve

$$\inf_{\underline{v_h}\in V_h}\ \sup_{q_h \in Q_h}\ \mu \int_\Omega |\underline{\underline{\varepsilon}}(\underline{v_h})|^2 dx - \int_\Omega \underline{f}\cdot \underline{v_h}\, dx - \int_\Omega q_h\ \mathrm{div}\, \underline{v_h}\, dx. \tag{1.5.31}$$

Let us suppose that the couple $V_h \times Q_h$ does not satisfy the stability conditions developed on Chap. 8 but that there exists a subspace Q_h^S of Q_h such that the couple $V_h \times Q_h^S$ is stable. If for some reason we want to use Q_h instead of Q_h^S we could retrieve stability by working with the problem

$$\inf_{\underline{v_h}\in V_h}\ \sup_{q_h \in Q_h}\ \mu \int_\Omega |\underline{\underline{\varepsilon}}(\underline{v_h})|^2 dx - \int_\Omega \underline{f}\cdot \underline{v_h}\, dx - \int_\Omega q_h\, \mathrm{div}\, \underline{v_h}\, dx$$
$$- \frac{\alpha}{2}\int_\Omega |q_h - P q_h|^2\, dx, \tag{1.5.32}$$

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where Pq_h stands for the projection of q_h on Q_h^S, which we suppose to be (easily) computable. This is now a stable formulation which, for μ_1 large, yields almost exactly the solution in $V_h \times Q_h^S$. It must be noted that the modified problem is meaningless for the continuous problem so that we have a different way of stabilising. However, the difference sometimes disappears when augmented versions are discretised.

1.6 Bibliographical Remarks

The purpose of this chapter was to present examples which will be used later as a standing ground for our development. It was not possible in such a context to consider every case. We already referred the reader to [166, 167] or [277] where the mathematical analysis of the problems selected here (and of many others problems) can be found in a unified setting. Advanced presentation of elasticity problems can be found in [295] or in [147]. For the Navier-Stokes equations, among the huge amount of literature, one may consult [278, 350, 363] or [159], and for electromagnetic problems the classical [273]. We also refer to more engineering-oriented presentations such as [53, 254, 271] and [387]. In particular, non-linear problems and their treatment are described in these references.

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